

Pseudo-differential Operators and Microlocal Analysis

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Abstract

Pseudo-differential operators give a more general framework for studying differential operators by examining them in terms of frequencies rather than positions. The theory allows for the approximate inversion (via a parametrix) of a broad class of operators and for tracking how irregularities in a solution arise and propagate. This work first presents the theory principally, then applies it to a specific operator combining slow diffusion and strong directional transport, a situation where standard methods fail. We show that the operator can nevertheless be approximately inverted in certain directions, and, in a more restricted setting, in all directions. We construct this approximate inverse explicitly and precisely identify the obstacle that imposes these restrictions.

Résumé

Les opérateurs pseudo-différentiels offrent un point de vue plus général pour étudier les opérateurs différentiels, en les examinant à travers les fréquences plutôt que les positions. La théorie permet d'inverser de manière approchée (au moyen d'une parametrix) une vaste classe d'opérateurs, et de suivre la façon dont les irrégularités d'une solution apparaissent et se propagent. Ce travail présente d'abord principalement la théorie, puis les applique à un opérateur particulier combinant une diffusion lente et un transport directionnel fort, une situation où les méthodes habituelles échouent. Nous montrons que l'opérateur peut néanmoins être inversé de façon approchée dans certaines directions, et, dans un cadre plus restreint, dans toutes les directions. Nous construisons cet inverse approché explicitement et identifions précisément l'obstruction qui impose ces restrictions.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



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1. Introduction

Partial differential equations (PDEs) lie at the heart of many problems in analysis, and they call for tools that can treat differential operators in a systematic and flexible way. Classical operators, such as the Laplacian or the wave operator, act locally: the value of the output at a point depends only on the input in a small neighbourhood of that point.

Many situations in mathematical analysis and mathematical physics, however, involve operators that sit somewhere between local and global. It was to meet this need that the theory of pseudo-differential operators (Ψ DO) first emerged (Trèves, 1980).

Loosely speaking, a pseudo-differential operator extends a differential operator by replacing the integer powers in its differentiation symbol with a more general function of the frequency variable. Once one allows symbols that are not polynomials, several powerful tools become available: one can invert elliptic operators approximately, build parametrices, and follow the way singularities behave under the operator (Zworski, 2012).

This final theme is the concern of microlocal analysis, which studies distributions and operators in both the position and the frequency variables at once. Building on the foundational work of Hörmander (1983), microlocal analysis gives a precise account of the wave front set of a distribution. With it, one can pinpoint exactly where, and in which directions, singularities appear and travel under a given operator (Grigis and Sjöstrand, 1994).

1.1 Historical Background

The story of pseudo-differential operators and microlocal analysis can be read as a single movement: from the study of singular integrals towards a geometric understanding of singularities in phase space.

The systematic treatment of singular integral equations in several variables goes back to Tricomi (1928), Giraud (1934) and Mikhlin (1948). Work in this direction gathered pace from 1936 onwards and was drawn together in Mikhlin's influential survey of 1948 (Mikhlin, 1948). There Mikhlin set out the central idea of the symbol of a singular integral, and showed that composing two such operators amounts to multiplying their symbols. In 1952, Calderón and Zygmund (1952) published a landmark memoir that placed the L^p theory of singular integral operators on firm ground. Five years later they turned to linear PDEs, where Calderón and Zygmund (1957) showed that the principal part of an elliptic differential operator P of order m can be written, up to lower-order terms, as $P = H\Lambda^m$, with $\Lambda = (-\Delta)^{1/2}$ and H a singular integral operator of order zero.

The conceptual ground for pseudo-differential operators was laid in the mid-1960s. In 1965, Kohn and Nirenberg (1965) and Hörmander (1965) brought singular integral operators and differential operators together within a single algebra. This unified viewpoint set aside the older kernel-based definition in favour of one built on the Fourier transform.

Microlocal analysis came into its own in the early 1970s, largely through the work of Sato et al. (1973). The change of viewpoint brought a key fact to light: PDEs become markedly easier to handle once the analysis is moved into phase space, that is, onto the cotangent bundle, rather than the underlying base manifold.

Around the same time, and again starting from the classical Fourier transform, Lars Hörmander introduced in 1971 the notion of the *wave front set* of a distribution (Hörmander, 1971). This gives a precise

description of both the points and the directions (covectors) at which a distribution fails to be smooth, read off from how it behaves at infinity in the frequency domain.

These ideas were taken considerably further in the 1980s. Masaki Kashiwara and Pierre Schapira carried the principles of microlocal analysis over to the general theory of sheaves (Kashiwara and Schapira, 1982; Schapira and Kashiwara-P, 1990). For this they introduced the “microsupport” of a sheaf, a closed conic subset of the cotangent bundle that records exactly the codirections in which the sheaf fails to propagate locally.

The original motivation for this sheaf-theoretic approach came from systems of linear PDEs, in the language of D-modules. Its main achievement was to show, in a strikingly clean form, that the microsupport of the complex of holomorphic solutions of a coherent D-module always sits inside the characteristic variety of the system.

1.2 Importance of the Theory

Pseudo-differential operators hold a central position in modern mathematics, drawing together several large areas of analysis. In the study of PDEs they offer a flexible and effective route to results on existence, uniqueness and regularity. The construction of a parametrix for an elliptic operator, that is, an approximate inverse modulo smoothing terms, rests on the symbolic calculus of pseudo-differential operators and stands among the most useful tools in elliptic regularity theory (Wong, 2014; Hörmander, 1983).

The reach of the theory extends well beyond this. In spectral theory, pseudo-differential operators are indispensable for understanding the asymptotic distribution of the eigenvalues of an elliptic operator, most famously through Weyl’s law (Hörmander, 1983). The symbolic structure of these operators is what makes it possible to relate the spectral behaviour of an operator to the geometry and analysis encoded in its principal symbol, and so to build deep links between analysis and geometry (Zworski, 2012).

The same ideas appear in mathematical physics. In quantum mechanics, the Weyl quantisation procedure assigns to observables, viewed as functions on phase space, self-adjoint operators on a Hilbert space; this procedure is, in essence, a pseudo-differential calculus, and it has prompted a great deal of mathematical work (Zworski, 2012). More broadly still, the microlocal point of view has proved invaluable in the analysis of wave propagation, scattering theory and hyperbolic equations (Grigis and Sjöstrand, 1994).

1.3 Organisation of the Work

This dissertation gives an account of the theory of pseudo-differential operators on $\mathbb{R}^n$, moving from the analytical foundations through to a non-trivial application in the non-elliptic setting.

Chapter 2 sets out the analytical prerequisites: multi-index notation, differentiation under the integral sign, convolution, tempered distributions, the Fourier transform on the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ and Partition of unity and Taylor’s formula. These tools underpin every construction that follows.

Chapter 3 develops the core calculus of pseudo-differential operators in the standard Kohn–Nirenberg class $S^m = S^m_{1,0}(\mathbb{R}^n \times \mathbb{R}^n)$ (Kohn and Nirenberg, 1965). We prove the asymptotic summation theorem, the composition formula for the product of two pseudo-differential operators, the construction of the formal adjoint, and the existence of a two-sided parametrix for elliptic operators. Throughout, full

proofs are given, with the emphasis on how the symbolic calculus turns operator-theoretic questions into algebraic ones.

Chapter 4 puts this calculus to work on a problem that lies outside the classical elliptic theory: the fractional Bessel potential with drift,

$$\mathcal{L} = (I - \Delta)^s + \nu(x) \cdot \nabla, \quad s \in \left(0, \frac{1}{2}\right),$$

in the supercritical regime. After establishing the failure of global ellipticity and proving a directional ellipticity estimate on a conic region of frequency space, we construct a two-sided parametrix for $\mathcal{L}$ in the Hörmander class $S_{2s, 1-2s}^{-2s}$ for $s \in (1/4, 1/2)$, and compute the formal L^2 -adjoint of $\mathcal{L}$ explicitly.

2. Preliminaries: Function spaces and Fourier transform

We establish the mathematical base required for the theory of pseudo-differential operators in this chapter. It is organised as follows: Section 2.1 introduces multi-index notation and basic differential operators; Section 2.2 discusses differentiation under the integral sign; Section 2.3 treats convolution and density; Section 2.4 on tempered distributions; Section 2.5 develops the Fourier transform on $\mathcal{S}(\mathbb{R}^n)$; and 2.6 introduce the theory of tempered distributions and A Partition of Unity with Taylor's Formula, respectively.

2.1 Notations and Differential Operators

We consider the n -dimensional Euclidean space

$$\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}, i = [1, n]\},$$

equipped with the standard Euclidean norm $|x| = (\sum_{j=1}^n x_j^2)^{1/2}$.

2.1.1 Multi-indices and Derivatives

2.1.2 Definition (Multi-index (Wong, 2014)). We define the *multi-index* $\alpha = (\alpha_1, \dots, \alpha_n)$ as an n -tuple of non-negative integers. And:

- **Length:** $|\alpha| = \sum_{j=1}^n \alpha_j$.
- **Factorial:** $\alpha! = \alpha_1! \alpha_2! \cdots \alpha_n!$.
- **Partial derivative:** $\partial^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n}$.
- **Differential operator:** $D_j = -i\partial_{x_j}$, $i^2 = -1$, and $D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$.

2.1.3 Lemma (Multi-index inequality (Wong, 2014, Exercise 1.5)). $\forall x \in \mathbb{R}^n$ and $\forall \alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$,

$$|x^\alpha| \leq |x|^{|\alpha|}.$$

Proof. Write $|x^\alpha| = \prod_{j=1}^n |x_j|^{\alpha_j}$. By the mean inequality applied in $\mathbb{R}^n$,

$$|x_j| \leq |x|, \forall j \in \{1, \dots, n\}.$$

Therefore

$$|x^\alpha| = \prod_{j=1}^n |x_j|^{\alpha_j} \leq \prod_{j=1}^n |x|^{\alpha_j} = |x|^{\sum_{j=1}^n \alpha_j} = |x|^{|\alpha|}.$$

□

2.1.4 Leibniz Formula and Polynomials

For f and $g \in C^\infty$, the Leibniz formula for multi-indices is given by:

$$D^\alpha(fg) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (D^\beta f)(D^{\alpha-\beta} g), \text{ where } \binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!}$$

A notable formula for the derivative of a reciprocal function is: for $f \in C^\infty(\mathbb{R}^n)$ with $f(x) \neq 0$,

$$D^\alpha \left(\frac{1}{f} \right) = \sum_{\substack{\alpha^{(1)} + \dots + \alpha^{(k)} = \alpha \\ |\alpha^{(j)}| \geq 1}} C_{\alpha^{(1)}, \dots, \alpha^{(k)}} \frac{(D^{\alpha^{(1)}} f) \dots (D^{\alpha^{(k)}} f)}{f^{k+1}}. \quad (2.1.1)$$

The sum is over all ordered partitions of α into $k \geq 1$ multi-indices $\alpha^{(j)}$ with $|\alpha^{(j)}| \geq 1$, and $C_{\alpha^{(1)}, \dots, \alpha^{(k)}}$ are integer constants (Wong, 2014, formula (1.4)).

2.1.5 Minkowski's Inequality

The integral form of Minkowski's inequality is a vital tool for $L^p(\mathbb{R}^n)$ estimates:

2.1.6 Theorem (Minkowski's Inequality; Integral Form (Wong, 2014, formula (1.5))). *Let $1 \leq p < \infty$ and $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$ (measurable function) such that $x \mapsto \int_{\mathbb{R}^n} |f(x, y)| dy$ belongs to $L^p(\mathbb{R}^n)$. Then*

$$\left(\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x, y) dy \right|^p dx \right)^{1/p} \leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x, y)|^p dx \right)^{1/p} dy. \quad (2.1.2)$$

2.2 Differentiation under the Integral Sign

2.2.1 Theorem (Differentiation under the Integral Sign (Wong, 2014)). *Let (Y, μ) be a measure space and let $f : \mathbb{R}^n \times Y \rightarrow \mathbb{C}$ (measurable function) satisfying the following conditions:*

1. $f(x, \cdot) \in L^1(Y)$ for all $x \in \mathbb{R}^n$;
2. $f(\cdot, y) \in C^\infty(\mathbb{R}^n)$ for μ -almost every $y \in Y$;
3. $\sup_{x \in \mathbb{R}^n} \int_Y |D_x^\alpha f(x, y)| d\mu(y) < \infty$ for every multi-index α .

Then the function F defined by $F(x) = \int_Y f(x, y) d\mu(y)$ on $C^\infty(\mathbb{R}^n)$ and

$$D^\alpha F(x) = \int_Y D_x^\alpha f(x, y) d\mu(y) \quad \text{for all } x \in \mathbb{R}^n.$$

2.2.2 Remark. Condition (3) requires a *uniform* L^1 bound on the derivatives with respect to x . This is stronger than what is strictly necessary: a dominated convergence argument shows that it suffices to have, for each compact set $K \subset \mathbb{R}^n$ and multi-index α , a function $g_{K, \alpha} \in L^1(Y)$ such that $|D_x^\alpha f(x, y)| \leq g_{K, \alpha}(y)$ for all $x \in K$ and μ -almost every $y \in Y$. The global uniformity in (3) is, however, well-adapted to the pseudo-differential operators on $\mathbb{R}^n$, where K ranges over all of $\mathbb{R}^n$.

2.2.3 Lemma. (Wong, 2014) Under the same hypotheses, if $x \mapsto D_x^\alpha f(x, y)$ is continuous for almost all y , then $x \mapsto \int_Y D_x^\alpha f(x, y) d\mu(y)$ is continuous on $\mathbb{R}^n$.

2.3 Convolution and Density

2.3.1 Definition (Schwartz space (Wong, 2014)). The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ is the vector space of all $\varphi \in C^\infty(\mathbb{R}^n)$ such that

$$\|\varphi\|_{\alpha, \beta} := \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \varphi(x)| < \infty$$

for all multi-indices α, β .

2.3.2 Definition (Convolution). For $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$, the convolution $(f * g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)dy$ exists and:

- $\|f * g\|_p \leq \|f\|_1 \|g\|_p$.
- If $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and $f \in L^p(\mathbb{R}^n)$, then $f * \varphi \in C^\infty(\mathbb{R}^n)$ and $\partial^\alpha(f * \varphi) = f * (\partial^\alpha \varphi)$.

2.3.3 Remark (Density (Wong, 2014)). Both the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ and the space of smooth functions with compact support $C_c^\infty(\mathbb{R}^n)$ are dense in $L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$.

2.4 Tempered Distributions

2.4.1 Definition (Convergence in $\mathcal{S}(\mathbb{R}^n)$ (Wong, 2014)). A sequence $\{\phi_j\} \subset \mathcal{S}(\mathbb{R}^n)$ is said to converge to zero in $\mathcal{S}(\mathbb{R}^n)$ if for all multi-indices α, β :

$$\sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \phi_j(x)| \rightarrow 0 \text{ as } j \rightarrow \infty$$

2.4.2 Definition (Tempered distribution (Wong, 2014)). A *tempered distribution* is a continuous linear functional $T : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$, i.e., T is linear and $T(\varphi_j) \rightarrow 0$ whenever $\varphi_j \rightarrow 0$ in $\mathcal{S}(\mathbb{R}^n)$. The space of tempered distributions is denoted $\mathcal{S}'(\mathbb{R}^n)$.

2.4.3 Proposition. (Wong, 2014)

- i. Any locally integrable function f satisfying $\int_{\mathbb{R}^n} \frac{|f(x)|}{(1+|x|)^N} dx < \infty$ for some $N \in \mathbb{N}$ defines a tempered distribution via $T_f(\varphi) = \int_{\mathbb{R}^n} f(x)\varphi(x) dx$.
- ii. The Fourier transform of $T \in \mathcal{S}'(\mathbb{R}^n)$ is defined by $\hat{T}(\varphi) = T(\hat{\varphi})$; one verifies that $\hat{T} \in \mathcal{S}'(\mathbb{R}^n)$.
- iii. $\hat{\hat{T}} = \tilde{T}$ (see Proposition 2.5.4).

2.5 Fourier Transform

2.5.1 Definition (Fourier Transform (Wong, 2014)). For $f \in L^1(\mathbb{R}^n)$, we define:

$$\hat{f}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} f(x) dx, \quad \xi \in \mathbb{R}^n$$

2.5.2 Proposition (Convolution theorem (Wong, 2014, Ch. 4)). Let $f, g \in L^1(\mathbb{R}^n)$. We have

$$\widehat{f * g}(\xi) = (2\pi)^{n/2} \hat{f}(\xi) \hat{g}(\xi). \quad (2.5.1)$$

Proof. By Fubini's theorem (applicable since $f, g \in L^1$):

$$\begin{aligned} \widehat{f * g}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} (f * g)(x) dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} \int_{\mathbb{R}^n} f(x-y)g(y) dy dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} g(y) \int_{\mathbb{R}^n} e^{-ix \cdot \xi} f(x-y) dx dy. \end{aligned}$$

Setting $z = x - y$ in the inner integral:

$$\begin{aligned}
 \widehat{f * g}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} g(y) \int_{\mathbb{R}^n} e^{-i(z+y)\cdot\xi} f(z) dz dy \\
 &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} g(y) e^{-iy\cdot\xi} \int_{\mathbb{R}^n} e^{-iz\cdot\xi} f(z) dz dy \\
 &= (2\pi)^{n/2} \underbrace{(2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz\cdot\xi} f(z) dz}_{=\hat{f}(\xi)} \underbrace{(2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iy\cdot\xi} g(y) dy}_{=\hat{g}(\xi)} \\
 &= (2\pi)^{n/2} \hat{f}(\xi) \hat{g}(\xi).
 \end{aligned}$$

□

2.5.3 Proposition (Fourier transform of polynomial multiplication (Wong, 2014)). For $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and every multi-index β ,

$$(D_\xi^\beta \hat{\varphi})(\xi) = \mathcal{F}[(-x)^\beta \varphi(x)](\xi), \quad (2.5.2)$$

where $D_\xi^\beta = (-i)^{|\beta|} \partial_\xi^\beta$ acts on the frequency variable ξ , and $(-x)^\beta = (-1)^{|\beta|} x^\beta$.

Proof. For $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and $|\beta| = 1$ (i.e., $\beta = e_j$, the j -th standard basis vector), we differentiate under the integral sign (justified by Theorem 2.2.1), since $|x_j e^{-ix\cdot\xi} \varphi(x)| \leq |x| |\varphi(x)|$ and $x\varphi \in L^1(\mathbb{R}^n)$ for $\varphi \in \mathcal{S}(\mathbb{R}^n)$:

$$\partial_{\xi_j} \hat{\varphi}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \partial_{\xi_j} (e^{-ix\cdot\xi}) \varphi(x) dx = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (-ix_j) e^{-ix\cdot\xi} \varphi(x) dx = \mathcal{F}[(-ix_j) \varphi](\xi).$$

Since $D_{\xi_j} = -i \partial_{\xi_j}$, we obtain $D_{\xi_j} \hat{\varphi}(\xi) = \mathcal{F}[(-x_j) \varphi](\xi)$. The general case $D_\xi^\beta \hat{\varphi} = \mathcal{F}[(-x)^\beta \varphi]$ follows by induction on $|\beta|$. □

2.5.4 Proposition ((Wong, 2014, pro 5.2)). Let $T \in \mathcal{S}'(\mathbb{R}^n)$. Define $\tilde{T}(\varphi) = T(\tilde{\varphi})$ for $\varphi \in \mathcal{S}(\mathbb{R}^n)$, where $\tilde{\varphi}(x) = \varphi(-x)$. Then

$$\hat{\hat{T}} = \tilde{T}. \quad (2.5.3)$$

Proof. For any $\varphi \in \mathcal{S}(\mathbb{R}^n)$, by the definition of the Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$:

$$\hat{\hat{T}}(\varphi) = \hat{T}(\hat{\varphi}) = T(\hat{\hat{\varphi}}).$$

By the Fourier inversion formula for Schwartz functions, $\hat{\hat{\varphi}}(x) = \varphi(-x) = \tilde{\varphi}(x)$. Hence

$$\hat{\hat{T}}(\varphi) = T(\tilde{\varphi}) = \tilde{T}(\varphi). \quad \square$$

2.5.5 Theorem (Fourier Inversion Formula (Wong, 2014, Theorem 4.4)). For every $\varphi \in \mathcal{S}(\mathbb{R}^n)$,

$$\varphi(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix\cdot\xi} \hat{\varphi}(\xi) d\xi, \quad x \in \mathbb{R}^n. \quad (2.5.4)$$

In other words, $(\hat{\varphi})^\vee = \varphi$, where $h^\vee(x) = h(-x)$ denotes the reflection of h .

Proof sketch. One first establishes (2.5.4) for the Gaussian $\varphi_0(x) = e^{-|x|^2/2}$ by direct computation. The general case follows by a density argument and the dominated convergence theorem; see (Wong, 2014, pp. 22-26) for the complete proof. □

2.5.6 Theorem (Plancherel's Theorem ([Wong, 2014](#), Theorem 4.5)). *The Fourier transform $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ extends uniquely to a unitary operator*

$$\mathcal{F} : L^2(\mathbb{R}^n) \longrightarrow L^2(\mathbb{R}^n),$$

satisfying $\|\hat{f}\|_{L^2} = \|f\|_{L^2}$ for all $f \in L^2(\mathbb{R}^n)$.

2.5.7 Proposition (Further properties of the Fourier transform ([Wong, 2014](#))). Let $f \in L^1(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$.

1. **Riemann-Lebesgue Lemma:** $\hat{f}$ is continuous and $\lim_{|\xi| \rightarrow \infty} \hat{f}(\xi) = 0$.
2. **Adjoint formula:** $\int_{\mathbb{R}^n} \hat{f}(x)g(x) dx = \int_{\mathbb{R}^n} f(x)\hat{g}(x) dx$ for all $g \in L^1(\mathbb{R}^n)$.
3. **Derivatives:** For $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and multi-index α , $\widehat{D^\alpha \varphi}(\xi) = \xi^\alpha \hat{\varphi}(\xi)$.

2.6 Partition of unity and Taylor's formula

2.6.1 Theorem. ([Wong, 2014](#)) *There is a sequence $\{\varphi_k\}_{k=0}^\infty$ of functions in $C_c^\infty(\mathbb{R}^n)$ such that*

- (i) $0 \leq \varphi_k(\xi) \leq 1$, $\xi \in \mathbb{R}^n$, $k = 0, 1, 2, \dots$,
- (ii) $\sum_{k=0}^\infty \varphi_k(\xi) = 1$, $\xi \in \mathbb{R}^n$,
- (iii) *at each $\xi \in \mathbb{R}^n$, at least one and at most three of the φ_k s are nonzero*
- (iv) $\text{supp}(\varphi_0) \subseteq \{\xi \in \mathbb{R}^n : |\xi| \leq 2\}$,
- (v) $\text{supp}(\varphi_k) \subseteq \{\xi \in \mathbb{R}^n : 2^{k-2} \leq |\xi| \leq 2^{k+1}\}$, $k = 1, 2, \dots$,
- (vi) *for each multi-index α , there is a constant $A_\alpha > 0$ such that*

$$\sup_{\xi \in \mathbb{R}^n} |(\partial^\alpha \varphi_k)(\xi)| \leq A_\alpha 2^{-k|\alpha|}, \quad k = 0, 1, 2, \dots$$

2.6.2 Theorem. ([Wong, 2014](#)) *Let $f \in C^\infty(\mathbb{R}^n)$. Then for all positive integers N ,*

$$\begin{aligned} & f(\xi + \eta) \\ &= \sum_{|\alpha| < N} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha + N \sum_{|\gamma| = N} \frac{\eta^\gamma}{\gamma!} \int_0^1 (1 - \theta)^{N-1} (\partial^\gamma f)(\xi + \theta\eta) d\theta \end{aligned} \quad (2.6.1)$$

for all $\xi, \eta \in \mathbb{R}^n$.

3. Construction and calculus of pseudo-differential operators

The foundational papers of [Kohn and Nirenberg \(1965\)](#) and [Hörmander \(1965\)](#) placed pseudo-differential operators among the most powerful and versatile tools in the modern study of partial differential equations. The idea behind them is simple: instead of viewing a differential operator as a chain of successive derivatives, one reads it as a multiplication in frequency space, an interpretation made possible by the Fourier transform.

Take any linear partial differential operator

$$P(x, D) := \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$$

on $\mathbb{R}^n$, whose coefficients a_α are functions on $\mathbb{R}^n$. Its **symbol** is defined by

$$p(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha. \quad (3.0.1)$$

Combining Propositions [2.5.7](#) and [2.5.5](#), one obtains, for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, the representation

$$(P(x, D)\varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} p(x, \xi) \hat{\varphi}(\xi) d\xi. \quad (3.0.2)$$

To see this, begin with the definition of $P(x, D)$ and apply, in turn, the identity $\widehat{D^\alpha \varphi} = \xi^\alpha \hat{\varphi}$ (Proposition [2.5.7](#)) together with the Fourier inversion formula:

$$\begin{aligned} (P(x, D)\varphi)(x) &= \sum_{|\alpha| \leq m} a_\alpha(x) (D^\alpha \varphi)(x) = \sum_{|\alpha| \leq m} a_\alpha(x) (\xi^\alpha \hat{\varphi}(\xi))^\vee(x) \\ &= \sum_{|\alpha| \leq m} a_\alpha(x) (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \xi^\alpha \hat{\varphi}(\xi) d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \underbrace{\left(\sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \right)}_{= p(x, \xi)} \hat{\varphi}(\xi) d\xi. \end{aligned}$$

Written in this form, the construction points to a natural generalisation: one may replace the polynomial symbol $p(x, \xi)$ by a far broader class of functions $\sigma(x, \xi)$, subject only to suitable growth conditions in ξ . The operators that arise in this way are the **pseudo-differential operators**.

3.1 Symbols and pseudo-differential operators

3.1.1 Definition (Class S^m , after [Wong, 2014](#), Def. 6.1, p. 32). Fix $m \in (-\infty, +\infty)$. We write S^m for the collection of all functions $\sigma(x, \xi) \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ with the following property: to each pair of multi-indices α and β there corresponds a constant $C_{\alpha, \beta} > 0$, depending only on α and β , such that

$$|(D_\xi^\alpha D_x^\beta \sigma)(x, \xi)| \leq C_{\alpha, \beta} (1 + |\xi|)^{m - |\alpha|}$$

holds for every $(x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n$. Any σ lying in $\bigcup_{m \in \mathbb{R}} S^m$ is called a *symbol*.

3.1.2 Definition (Pseudo-differential operator T_σ , after Wong, 2014, Def. 6.2, p. 32). Given a symbol σ , the associated **pseudo-differential operator** T_σ is given by

$$(T_\sigma \varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma(x, \xi) \hat{\varphi}(\xi) d\xi, \quad \varphi \in \mathcal{S}(\mathbb{R}^n). \quad (3.1.1)$$

3.2 Fundamental examples

3.2.1 Example. Let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$ be a linear partial differential operator on $\mathbb{R}^n$. Suppose every coefficient a_α is smooth and has bounded derivatives of all orders. Then the associated polynomial $p(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha$ belongs to S^m , so that $P(x, D)$ is a pseudo-differential operator.

Proof. Let

$$P(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha,$$

where $a_\alpha \in C^\infty(\mathbb{R}^n)$ with all partial derivatives bounded on $\mathbb{R}^n$.

We prove that $p \in S^m$. For multi-indices $\beta, \gamma \in \mathbb{N}^n$, we compute

$$D_\xi^\beta D_x^\gamma p(x, \xi) = D_\xi^\beta D_x^\gamma \left(\sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \right) = \sum_{|\alpha| \leq m} D_\xi^\beta (D_x^\gamma a_\alpha(x) \xi^\alpha) = \sum_{|\alpha| \leq m} (D_x^\gamma a_\alpha(x)) (D_\xi^\beta \xi^\alpha).$$

Taking absolute values,

$$\left| D_\xi^\beta D_x^\gamma p(x, \xi) \right| \leq \sum_{|\alpha| \leq m} |D_x^\gamma a_\alpha(x)| \left| D_\xi^\beta \xi^\alpha \right|.$$

Since all coefficients a_α are C^∞ with bounded derivatives, there exist constants $C_{\alpha, \gamma} > 0$ such that $|D_x^\gamma a_\alpha(x)| \leq C_{\alpha, \gamma}$ for all $x \in \mathbb{R}^n$. Moreover,

$$D_\xi^\beta \xi^\alpha = \begin{cases} \frac{\alpha!}{(\alpha - \beta)!} \xi^{\alpha - \beta}, & \text{if } \beta \leq \alpha, \\ 0, & \text{otherwise,} \end{cases}$$

so that $\left| D_\xi^\beta \xi^\alpha \right| \leq C_{\alpha, \beta} |\xi|^{|\alpha| - |\beta|}$ for some constant $C_{\alpha, \beta} > 0$.

Therefore,

$$\begin{aligned} \left| D_\xi^\beta D_x^\gamma p(x, \xi) \right| &\leq \sum_{|\alpha| \leq m} C_{\alpha, \gamma} C_{\alpha, \beta} |\xi|^{|\alpha| - |\beta|} \leq \sum_{|\alpha| \leq m} C_{\alpha, \gamma, \beta} (1 + |\xi|)^{|\alpha| - |\beta|} \\ &\leq \left(\sum_{|\alpha| \leq m} C_{\alpha, \gamma, \beta} \right) (1 + |\xi|)^{m - |\beta|}. \end{aligned}$$

Note that when $|\beta| > |\alpha|$, one has $D_\xi^\beta \xi^\alpha = 0$, so only the case $|\beta| \leq |\alpha|$ contributes, in which $|\alpha| - |\beta| \geq 0$ and the bound $|\xi|^{|\alpha| - |\beta|} \leq (1 + |\xi|)^{|\alpha| - |\beta|}$ holds trivially.

Setting $C_{\beta, \gamma} := \sum_{|\alpha| \leq m} C_{\alpha, \gamma, \beta}$, we obtain

$$\left| D_\xi^\beta D_x^\gamma p(x, \xi) \right| \leq C_{\beta, \gamma} (1 + |\xi|)^{m - |\beta|}$$

for all $(x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n$, which shows that $p \in S^m$. □

3.3 Asymptotic expansion

An important notion in the theory of pseudo-differential operators is the asymptotic expansion of a symbol.

3.3.1 Definition. Asymptotic expansion - (Wong, 2014, Def. 6.9, p. 36). Let $\sigma \in S^m$. Suppose we can find $\sigma_j \in S^{m_j}$, where $m = m_0 > m_1 > m_2 > \cdots > m_j \rightarrow -\infty$ as $j \rightarrow \infty$, such that

$$\sigma - \sum_{j=0}^{N-1} \sigma_j \in S^{m_N}, \text{ for every positive integer } N.$$

Then we call $\sum_{j=0}^{\infty} \sigma_j$ an *asymptotic expansion* of σ and write $\sigma \sim \sum_{j=0}^{\infty} \sigma_j$.

3.3.2 Theorem. (Wong, 2014, Thm. 6.10, p. 36). Let $m_0 > m_1 > m_2 > \cdots > m_j \rightarrow -\infty$ as $j \rightarrow \infty$. Suppose $\sigma_j \in S^{m_j}$. Then there exists a symbol $\sigma \in S^{m_0}$ such that

$$\sigma \sim \sum_{j=0}^{\infty} \sigma_j.$$

Moreover, if τ is another symbol with the same asymptotic expansion, then $\sigma - \tau \in \bigcap_{k \in \mathbb{R}} S^k$.

Proof. Let $\psi \in C^\infty(\mathbb{R}^n)$ be a function such that

$$0 \leq \psi(\xi) \leq 1, \quad \xi \in \mathbb{R}^n, \quad \psi(\xi) = 0, \quad |\xi| \leq 1, \quad \psi(\xi) = 1, \quad |\xi| \geq 2,$$

and let $\{\varepsilon_j\}_{j \geq 0}$ be a sequence of positive numbers such that

$$1 > \varepsilon_0 > \varepsilon_1 > \varepsilon_2 > \cdots > \varepsilon_j \rightarrow 0 \quad \text{as } j \rightarrow \infty.$$

We define the function σ on $\mathbb{R}^n \times \mathbb{R}^n$ by

$$\sigma(x, \xi) = \sum_{j=0}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi), \quad x, \xi \in \mathbb{R}^n. \quad (3.3.1)$$

But for any $(x_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^n$, there exist a neighborhood U of (x_0, ξ_0) and a positive integer N such that

$$\psi(\varepsilon_j \xi) \sigma_j(x, \xi) = 0, \quad \forall (x, \xi) \in U, \quad \forall j > N,$$

since $\varepsilon_j \rightarrow 0$ guarantees that $\varepsilon_j |\xi| \leq 1$ for j large enough on a neighborhood of ξ_0 , and ψ vanishes on $\{|\xi| \leq 1\}$. Hence $\sigma \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$.

Furthermore, for any nonzero multi-index α ,

$$\partial_\xi^\alpha [\psi(\varepsilon \xi)] = \varepsilon^{|\alpha|} (\partial^\alpha \psi)(\varepsilon \xi) = 0 \quad \text{if } \varepsilon |\xi| \leq 1 \text{ or } \varepsilon |\xi| \geq 2,$$

and

$$|\partial_\xi^\alpha \psi(\varepsilon \xi)| \leq C_\alpha \varepsilon^{|\alpha|}, \quad 1 \leq \varepsilon |\xi| \leq 2,$$

where $C_\alpha = \sup_{\xi \in \mathbb{R}^n} |\partial_\xi^\alpha \psi(\xi)|$. On $\{1 \leq \varepsilon|\xi| \leq 2\}$ we have $\frac{1}{|\xi|} \leq \varepsilon \leq \frac{2}{|\xi|}$, so $\varepsilon \leq \frac{4}{1+|\xi|}$ whenever $|\xi| \geq 1$, and therefore

$$|\partial_\xi^\alpha \psi(\varepsilon\xi)| \leq C'_\alpha (1+|\xi|)^{-|\alpha|}, \quad \xi \in \mathbb{R}^n, \quad (3.3.2)$$

with $C'_\alpha = 4^{|\alpha|} C_\alpha$, the estimate being trivially valid for $\alpha = 0$ and outside $\{1 \leq \varepsilon|\xi| \leq 2\}$.

Now, for any multi-indices a, b , by Leibniz's formula and (3.3.2),

$$\begin{aligned} |D_\xi^a D_x^b [\psi(\varepsilon_j \xi) \sigma_j(x, \xi)]| &= \left| \sum_{\gamma \leq a} \binom{a}{\gamma} (D_\xi^\gamma \psi(\varepsilon_j \xi)) (D_\xi^{a-\gamma} D_x^b \sigma_j)(x, \xi) \right| \\ &\leq \sum_{\gamma \leq a} \binom{a}{\gamma} C'_\gamma (1+|\xi|)^{-|\gamma|} C_{j,a,b,\gamma} (1+|\xi|)^{m_j-|a|+|\gamma|} \\ &\leq C'_{j,a,b} (1+|\xi|)^{m_j-|a|}, \text{ where } C'_{j,a,b} = \sum_{\gamma \leq a} \binom{a}{\gamma} C'_\gamma C_{j,a,b,\gamma}. \end{aligned}$$

Moreover, on the support of $\psi(\varepsilon_j \xi)$ we have $\varepsilon_j |\xi| \geq 1$, hence $\varepsilon_j \geq \frac{1}{1+|\xi|}$. So, since the constants $C'_{j,a,b}$ for $|a| + |b| \leq j$ are finite in number, we can choose ε_j recursively so small that

$$C'_{j,a,b} (1+|\xi|)^{-1} \leq C'_{j,a,b} \varepsilon_j \leq 2^{-j} \quad \text{for every } a, b \text{ with } |a| + |b| \leq j. \quad (3.3.3)$$

Fix multi-indices a, b and take j_0 so large that $j_0 \geq |a+b|$ and $m_{j_0} + 1 \leq m_0$. Write

$$\sigma(x, \xi) = \sum_{j=0}^{j_0-1} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) + \sum_{j=j_0}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) = I(x, \xi) + J(x, \xi).$$

The sum $I(x, \xi)$ is finite, and each term $\psi(\varepsilon_j \xi) \sigma_j$ belongs to $S^{m_j} \subset S^{m_0}$, so $I \in S^{m_0}$. For $J(x, \xi)$, combining (3.3.3) with the termwise estimate above and the inequality $j_0 \geq |a+b|$ and $m_{j_0} + 1 \leq m_0$ on the support of $\psi(\varepsilon_j \xi)$,

$$\begin{aligned} |D_\xi^a D_x^b J(x, \xi)| &\leq \sum_{j=j_0}^{\infty} C'_{j,a,b} (1+|\xi|)^{m_j-|a|} \leq \sum_{j=j_0}^{\infty} 2^{-j} (1+|\xi|)^{m_j+1-|a|} \\ &\leq \sum_{j=j_0}^{\infty} 2^{-j} (1+|\xi|)^{m_0-|a|} = 2^{-j_0+1} (1+|\xi|)^{m_0-|a|}. \end{aligned}$$

Hence $J \in S^{m_0}$, so $\sigma = I + J \in S^{m_0}$.

Next we need to show that $\sigma - \sum_{j=0}^{N-1} \sigma_j \in S^{m_N}$ for every positive integer N . We have

$$\begin{aligned} \sigma(x, \xi) - \sum_{j=0}^{N-1} \sigma_j(x, \xi) &= \sum_{j=0}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) - \sum_{j=0}^{N-1} \sigma_j(x, \xi) \\ &= \sum_{j=0}^{N-1} (\psi(\varepsilon_j \xi) - 1) \sigma_j(x, \xi) + \sum_{j=N}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi). \end{aligned}$$

We have that $\text{supp}(\psi(\varepsilon_j \cdot) - 1) \subseteq \{\xi \in \mathbb{R}^n : |\xi| \leq 2/\varepsilon_j\} =: A_j$, and A_j is bounded, so for each $j \in \{0, \dots, N-1\}$ the function $\xi \mapsto (\psi(\varepsilon_j \xi) - 1) \sigma_j(x, \xi)$ has compact support in ξ . Hence

$$\sum_{j=0}^{N-1} (\psi(\varepsilon_j \xi) - 1) \sigma_j(x, \xi) \in \bigcap_{k \in \mathbb{R}} S^k \subset S^{m_N},$$

and, repeating the previous argument with j_0 replaced by N (and using $m_j \leq m_N$ for $j \geq N$),

$$\sum_{j=N}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) \in S^{m_N}. \text{ Therefore } \sigma - \sum_{j=0}^{N-1} \sigma_j \in S^{m_N},$$

which establishes $\sigma \sim \sum_{j=0}^{\infty} \sigma_j$.

Finally, if τ is another symbol such that $\tau \sim \sum_{j=0}^{\infty} \sigma_j$, then for every positive integer N ,

$$\sigma - \tau = \left(\sigma - \sum_{j=0}^{N-1} \sigma_j \right) - \left(\tau - \sum_{j=0}^{N-1} \sigma_j \right) \in S^{m_N}.$$

Since $m_N \rightarrow -\infty$, it follows that $\sigma - \tau \in \bigcap_{m \in \mathbb{R}} S^m$. $\square$

3.4 Product of two pseudo-differential operators

A main result establishes that the product (or composition) of two pseudo-differential operators is again a pseudo-differential operator, and provides an asymptotic formula for the symbol of the product.

3.4.1 Lemma. Let $R_{N_1}(x, z, \xi)$ be the function given by (3.4.25). Then for all multi-indices α, β and γ , there exists a constant $C_{\alpha, \beta, \gamma} > 0$ such that

$$|(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi)| \leq C_{\alpha, \beta, \gamma} \left[\sum_{\gamma' \leq \gamma} (1 + |z|)^{N_1 - |\gamma'|} \right] (1 + |\xi|)^{m_2 - |\beta|} \quad (3.4.1)$$

for all $x, z, \xi \in \mathbb{R}^n$. In particular,

$$|(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi)| \leq C'_{\alpha, \beta, \gamma} (1 + |z|)^{N_1} (1 + |\xi|)^{m_2 - |\beta|}. \quad (3.4.2)$$

Proof.

$$R_{N_1}(x, z, \xi) = N_1 \sum_{|\mu|=N_1} \frac{(-z)^\mu}{\mu!} \int_0^1 (1 - \theta)^{N_1 - 1} (\partial_x^\mu \tau)(x - \theta z, \xi) d\theta \quad (3.4.3)$$

for all $x, z, \xi \in \mathbb{R}^n$.

Applying the chain rule to $\tau(x - \theta z, \xi)$ in the variable x gives $\partial_x^\alpha [\tau(x - \theta z, \xi)] = (\partial_x^\alpha \tau)(x - \theta z, \xi)$, so that

$$(\partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi) = N_1 \sum_{|\mu|=N_1} \frac{(-z)^\mu}{\mu!} \int_0^1 (1 - \theta)^{N_1 - 1} (\partial_x^{\mu + \alpha} \partial_\xi^\beta \tau)(x - \theta z, \xi) d\theta. \quad (3.4.4)$$

$$f(z) := (-z)^\mu, \quad g(z) := (\partial_x^{\mu + \alpha} \partial_\xi^\beta \tau)(x - \theta z, \xi).$$

Leibniz's formula gives

$$\begin{aligned} \partial_z^\gamma (fg) &= \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} (\partial_z^{\gamma'} f) (\partial_z^{\gamma - \gamma'} g). \\ \partial_z^{\gamma'} [(-z)^\mu] &= \begin{cases} (-1)^{|\mu|} \frac{\mu!}{(\mu - \gamma')!} z^{\mu - \gamma'}, & \text{if } \gamma' \leq \mu, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (3.4.5)$$

For $\partial_z^{\gamma-\gamma'} g$: by the chain rule, $\partial_{z_j}[\tau(x - \theta z, \xi)] = -\theta (\partial_{x_j} \tau)(x - \theta z, \xi)$, hence iterating,

$$\partial_z^{\gamma-\gamma'} g(z) = (-\theta)^{|\gamma-\gamma'|} (\partial_x^{\mu+\alpha+\gamma-\gamma'} \partial_\xi^\beta \tau)(x - \theta z, \xi). \quad (3.4.6)$$

Combining (3.4.5) and (3.4.6) in Leibniz's formula and integrating, we obtain

$$\begin{aligned} (\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi) &= N_1 \sum_{|\mu|=N_1} \sum_{\substack{\gamma' \leq \gamma \\ \gamma' \leq \mu}} \binom{\gamma}{\gamma'} \frac{(-1)^{|\mu|+|\gamma-\gamma'|}}{(\mu - \gamma')!} z^{\mu-\gamma'} \\ &\quad \times \int_0^1 (1-\theta)^{N_1-1} \theta^{|\gamma-\gamma'|} (\partial_x^{\mu+\alpha+\gamma-\gamma'} \partial_\xi^\beta \tau)(x - \theta z, \xi) d\theta. \end{aligned} \quad (3.4.7)$$

Since $\tau \in S^{m_2}$, yields, for each multi-index $v = \mu + \alpha + \gamma - \gamma'$, a constant $C_{v,\beta} > 0$ such that

$$|(\partial_x^\sigma \partial_\xi^\beta \tau)(y, \xi)| \leq C_{v,\beta} (1 + |\xi|)^{m_2-|\beta|} \quad \text{for all } y, \xi \in \mathbb{R}^n. \quad (3.4.8)$$

Applied at $y = x - \theta z$, the right-hand side of (3.4.8) does not depend on θ , x , or z .

Furthermore, $|z^{\mu-\gamma'}| \leq |z|^{|\mu|-|\gamma'|} = |z|^{N_1-|\gamma'|}$, and the Beta integral

$$\int_0^1 (1-\theta)^{N_1-1} \theta^{|\gamma-\gamma'|} d\theta = B(N_1, |\gamma-\gamma'|+1)$$

is a finite positive constant.

Taking absolute values in (3.4.7) and using the bounds above,

$$|(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi)| \leq N_1 \sum_{|\mu|=N_1} \sum_{\substack{\gamma' \leq \gamma \\ \gamma' \leq \mu}} \binom{\gamma}{\gamma'} \frac{C_{\mu+\alpha+\gamma-\gamma',\beta}}{(\mu - \gamma')!} B(N_1, |\gamma-\gamma'|+1) |z|^{N_1-|\gamma'|} (1 + |\xi|)^{m_2-|\beta|}.$$

Setting

$$C_{\alpha,\beta,\gamma} := N_1 \sum_{|\mu|=N_1} \sum_{\substack{\gamma' \leq \gamma \\ \gamma' \leq \mu}} \binom{\gamma}{\gamma'} \frac{C_{\mu+\alpha+\gamma-\gamma',\beta}}{(\mu - \gamma')!} B(N_1, |\gamma-\gamma'|+1)$$

(a finite sum over a finite set of multi-indices), we obtain

$$|(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi)| \leq C_{\alpha,\beta,\gamma} \left[\sum_{\gamma' \leq \gamma} |z|^{N_1-|\gamma'|} \right] (1 + |\xi|)^{m_2-|\beta|}. \quad (3.4.9)$$

Replacing $|z|^{N_1-|\gamma'|}$ by the larger quantity $(1 + |z|)^{N_1-|\gamma'|}$ yields (3.4.1).

For each $\gamma' \leq \gamma$ we have $|\gamma'| \leq |\gamma|$, hence $(1 + |z|)^{N_1-|\gamma'|} \leq (1 + |z|)^{N_1}$. The number of terms in the sum $\sum_{\gamma' \leq \gamma}$ is bounded by $\prod_{j=1}^n (\gamma_j + 1)$, a constant depending only on γ . Therefore there exists a constant $C'_{\alpha,\beta,\gamma} > 0$ such that

$$|(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1})(x, z, \xi)| \leq C'_{\alpha,\beta,\gamma} (1 + |z|)^{N_1} (1 + |\xi|)^{m_2-|\beta|},$$

which is (3.4.2). □

3.4.2 Lemma. For every nonnegative integer M , there exists a constant $C_{\alpha,\beta,M} > 0$ such that

$$\left| (\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)})(x, \xi) \right| \leq C_{\alpha,\beta,M} (1 + |\xi|)^{m_2 - 2M} 2^{(m_1 + 2M - N_1)k}$$

for all $(x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n$ and every $k \geq 0$.

Proof.

$$T_{N_1}^{(k)}(x, \xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \xi} K_k(x, z) R_{N_1}(x, z, \xi) dz,$$

where K_k is defined by

$$K_k(x, z) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{iz \cdot \xi} \sigma_k(x, \xi) d\xi, \quad x, z \in \mathbb{R}^n, \quad (3.4.10)$$

and R_{N_1} is given by (3.4.25). Let α, β be arbitrary multi-indices, and let M be any nonnegative integer. We aim to estimate $\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)}$.

$$(\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)})(x, \xi) = (2\pi)^{-n/2} \sum_{\substack{\beta' \leq \beta \\ \alpha' \leq \alpha}} \binom{\beta}{\beta'} \binom{\alpha}{\alpha'} \int_{\mathbb{R}^n} (-iz)^{\beta - \beta'} e^{-iz \cdot \xi} (\partial_x^{\alpha'} K_k)(x, z) (\partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi) dz, \quad (3.4.11)$$

since $\partial_\xi^{\beta - \beta'} (e^{-iz \cdot \xi}) = (-iz)^{\beta - \beta'} e^{-iz \cdot \xi}$.

The key idea is to convert decay in ξ into integrability in z . For any nonnegative integer M and any $\xi \neq 0$, we use the identity

$$e^{-iz \cdot \xi} = (-|\xi|^2)^{-M} \Delta_z^M e^{-iz \cdot \xi}, \quad (3.4.12)$$

where $\Delta_z = \sum_{j=1}^n \partial_{z_j}^2$. Indeed, $\Delta_z e^{-iz \cdot \xi} = -|\xi|^2 e^{-iz \cdot \xi}$, hence $\Delta_z^M e^{-iz \cdot \xi} = (-|\xi|^2)^M e^{-iz \cdot \xi}$, which gives (3.4.12).

Substituting (3.4.12) into each summand of (3.4.11) and performing $2M$ integrations by parts in z (the boundary terms vanish

because $K_k(x, \cdot) \in \mathcal{S}(\mathbb{R}^n)$ (so all boundary terms vanish), we obtain

$$\int (-iz)^{\beta - \beta'} e^{-iz \cdot \xi} (\partial_x^{\alpha'} K_k)(x, z) (\partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi) dz = (-|\xi|^2)^{-M} \int e^{-iz \cdot \xi} \Delta_z^M [(-iz)^{\beta - \beta'} (\partial_x^{\alpha'} K_k)(x, z) (\partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi)] dz. \quad (3.4.13)$$

$$\Delta_z^M [(-iz)^{\beta - \beta'} (\partial_x^{\alpha'} K_k)(x, z) (\partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi)] = \sum_{|\gamma_1| + |\gamma_2| + |\gamma_3| = 2M} c_{\gamma_1, \gamma_2, \gamma_3} \partial_z^{\gamma_1} [(-iz)^{\beta - \beta'}] \times (\partial_z^{\gamma_2} \partial_x^{\alpha'} K_k)(x, z) (\partial_z^{\gamma_3} \partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi),$$

where the $c_{\gamma_1, \gamma_2, \gamma_3}$ are combinatorial constants. $\partial_z^{\gamma_1} [(-iz)^{\beta - \beta'}]$ is a polynomial in z of degree $|\beta - \beta'| - |\gamma_1|$ (zero if $|\gamma_1| > |\beta - \beta'|$), so there exists a constant C such that

$$\left| \partial_z^{\gamma_1} [(-iz)^{\beta - \beta'}] \right| \leq C(1 + |z|)^{|\beta - \beta'| - |\gamma_1|}. \quad (3.4.14)$$

By Lemma 3.4.1 applied to R_{N_1} with the multi-indices $(\alpha - \alpha', \beta', \gamma_3)$, there exists a constant $C_{\alpha - \alpha', \beta', \gamma_3} > 0$ such that

$$\left| (\partial_z^{\gamma_3} \partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi) \right| \leq C_{\alpha - \alpha', \beta', \gamma_3} |z|^{N_1 - |\gamma_3|} (1 + |\xi|)^{m_2 - |\beta'|} \quad (3.4.15)$$

for all $x, z, \xi \in \mathbb{R}^n$, with the convention that the right-hand side vanishes when $|\gamma_3| > N_1$.

Combining (3.4.13), (3.4.14), and (3.4.15), each summand in $\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)}$ is bounded in absolute value by

$$|\xi|^{-2M} (1 + |\xi|)^{m_2 - |\beta'|} C \int_{\mathbb{R}^n} (1 + |z|)^{|\beta - \beta'| - |\gamma_1|} \left| (\partial_z^{\gamma_2} \partial_x^{\alpha'} K_k)(x, z) \right| |z|^{N_1 - |\gamma_3|} dz.$$

Since $|\gamma_1| + |\gamma_2| + |\gamma_3| = 2M$, the total power of z inside the integral is bounded by

$$|\beta - \beta'| - |\gamma_1| + N_1 - |\gamma_3| = |\beta - \beta'| + N_1 - (|\gamma_1| + |\gamma_3|),$$

and $(1 + |z|)^{|\beta - \beta'| - |\gamma_1|} |z|^{N_1 - |\gamma_3|} \leq (1 + |z|)^{N_1 + |\beta - \beta'| - |\gamma_1| - |\gamma_3|}$. In particular, there exists a nonnegative integer $\tilde{N}$ (depending on α, β, M) such that

$$(1 + |z|)^{|\beta - \beta'| - |\gamma_1|} |z|^{N_1 - |\gamma_3|} \leq (1 + |z|)^{\tilde{N}}.$$

For every nonnegative integer $\tilde{N}$ and every multi-index (γ_2, α') , there is a constant $A = A(\alpha', \gamma_2, \tilde{N}) > 0$ such that

$$\int_{\mathbb{R}^n} (1 + |z|)^{\tilde{N}} \left| (\partial_z^{\gamma_2} \partial_x^{\alpha'} K_k)(x, z) \right| dz \leq A 2^{(m_1 + |\gamma_2| - \tilde{N})k} \leq A 2^{(m_1 + 2M - \tilde{N})k}, \quad (3.4.16)$$

using $|\gamma_2| \leq 2M$.

$$\left| (\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)})(x, \xi) \right| \leq C'_{\alpha, \beta, M} |\xi|^{-2M} (1 + |\xi|)^{m_2} 2^{(m_1 + 2M - N_1)k} \quad (3.4.17)$$

for $\xi \neq 0$. For $|\xi| \geq 1$, we have $|\xi|^{-2M} \leq 2^{2M} (1 + |\xi|)^{-2M}$, so (3.4.17) give

$$\left| (\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)})(x, \xi) \right| \leq C''_{\alpha, \beta, M} (1 + |\xi|)^{m_2 - 2M} 2^{(m_1 + 2M - N_1)k}.$$

For $|\xi| \leq 1$, we estimate $\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)}$ directly from (3.4.11) without integration by parts. The integrand in each summand is bounded by

$$|z|^{|\beta - \beta'|} \left| (\partial_x^{\alpha'} K_k)(x, z) \right| \left| (\partial_x^{\alpha - \alpha'} \partial_\xi^{\beta'} R_{N_1})(x, z, \xi) \right| \leq C (1 + |z|)^{|\beta - \beta'| + N_1} \left| (\partial_x^{\alpha'} K_k)(x, z) \right| (1 + |\xi|)^{m_2 - |\beta'|},$$

by Lemma 3.4.1 (with $\gamma_3 = 0$). Since $(1 + |\xi|)^{m_2 - |\beta'|} \leq 2^{|m_2| + |\beta|}$ on the bounded set $|\xi| \leq 1$, gives

$$\left| (\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)})(x, \xi) \right| \leq C'''_{\alpha, \beta} 2^{(m_1 + |\alpha| - N_1 - |\beta|)k} \leq C'''_{\alpha, \beta} 2^{(m_1 + 2M - N_1)k} (1 + |\xi|)^{m_2 - 2M}$$

(the last inequality holds for $|\xi| \leq 1$ on the right-hand side since $(1 + |\xi|)^{m_2 - 2M} \geq 2^{-|m_2 - 2M|}$). Taking $C_{\alpha, \beta, M} = \max\{C''_{\alpha, \beta, M}, C'''_{\alpha, \beta}\}$ gives

$$\left| (\partial_x^\alpha \partial_\xi^\beta T_{N_1}^{(k)})(x, \xi) \right| \leq C_{\alpha, \beta, M} (1 + |\xi|)^{m_2 - 2M} 2^{(m_1 + 2M - N_1)k} \quad (3.4.18)$$

for all $x, \xi \in \mathbb{R}^n$ and all $k = 0, 1, 2, \dots$, which is the conclusion of Lemma 3.4.2. $\square$

3.4.3 Theorem. Let $\sigma \in S^{m_1}$ and $\tau \in S^{m_2}$. Then the product $T_\sigma T_\tau$ of the pseudo-differential operators T_σ and T_τ is again a pseudo-differential operator T_λ , where $\lambda \in S^{m_1+m_2}$ and has the asymptotic expansion

$$\lambda \sim \sum_{\mu} \frac{(-i)^{|\mu|}}{\mu!} \left(\partial_{\xi}^{\mu} \sigma \right) \left(\partial_x^{\mu} \tau \right). \quad (3.4.19)$$

Here, 3.4.3 means that $\lambda - \sum_{|\mu| < N} \frac{(-i)^{|\mu|}}{\mu!} \left(\partial_{\xi}^{\mu} \sigma \right) \left(\partial_x^{\mu} \tau \right)$ is a symbol in $S^{m_1+m_2-N}$ for every positive integer N .

Proof. Theorem 3.4.3 Let us first examine how to proceed formally. For any $\varphi \in \mathcal{S}$, by the definition of a pseudo-differential operator,

$$(T_\sigma \varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma(x, \xi) \widehat{\varphi}(\xi) d\xi, \quad x \in \mathbb{R}^n.$$

Let $\{\varphi_k\}_{k=0}^{\infty}$ be the partition of unity in the ξ variable constructed in Theorem 2.6.1, and set

$$\sigma_k(x, \xi) := \sigma(x, \xi) \varphi_k(\xi), \quad x, \xi \in \mathbb{R}^n. \quad (3.4.20)$$

Each σ_k has compact support in ξ . Since $\sigma = \sum_{k=0}^{\infty} \sigma_k$, Fubini's theorem yields

$$(T_\sigma \varphi)(x) = \sum_{k=0}^{\infty} (T_{\sigma_k} \varphi)(x),$$

and the convergence of the series can be shown to be absolute and uniform on $\mathbb{R}^n$.

By the definition of a pseudo-differential operator and the Fourier transform,

$$\begin{aligned} (T_{\sigma_k} T_\tau \varphi)(x) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma_k(x, \xi) \widehat{T_\tau \varphi}(\xi) d\xi \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y) \cdot \xi} \sigma_k(x, \xi) (T_\tau \varphi)(y) dy d\xi. \end{aligned}$$

The compact support of σ_k in ξ justifies the interchange of integrals by Fubini's theorem. Set

$$K_k(x, z) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{iz \cdot \xi} \sigma_k(x, \xi) d\xi, \quad x, z \in \mathbb{R}^n. \quad (3.4.21)$$

Then

$$(T_{\sigma_k} T_\tau \varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} K_k(x, x-y) (T_\tau \varphi)(y) dy. \quad (3.4.22)$$

Inserting the definition of $T_\tau \varphi$ into (3.4.22) and using Fubini's theorem once more,

$$\begin{aligned} (T_{\sigma_k} T_\tau \varphi)(x) &= (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{iy \cdot \eta} \tau(y, \eta) K_k(x, x-y) \widehat{\varphi}(\eta) dy d\eta \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \eta} \lambda_k(x, \eta) \widehat{\varphi}(\eta) d\eta, \end{aligned}$$

where, after the change of variable $z = x - y$,

$$\lambda_k(x, \xi) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \xi} K_k(x, z) \tau(x-z, \xi) dz. \quad (3.4.23)$$

Therefore $T_{\sigma_k} T_\tau = T_{\lambda_k}$.

By Taylor's formula with integral remainder (Theorem 2.6.2) applied to the function $z \mapsto \tau(x - z, \xi)$ around $z = 0$, for every positive integer N_1 ,

$$\tau(x - z, \xi) = \sum_{|\mu| < N_1} \frac{(-z)^\mu}{\mu!} (\partial_x^\mu \tau)(x, \xi) + R_{N_1}(x, z, \xi), \quad (3.4.24)$$

where

$$R_{N_1}(x, z, \xi) = N_1 \sum_{|\mu|=N_1} \frac{(-z)^\mu}{\mu!} \int_0^1 (1 - \theta)^{N_1-1} (\partial_x^\mu \tau)(x - \theta z, \xi) d\theta. \quad (3.4.25)$$

Substituting (3.4.24) into (3.4.23),

$$\lambda_k(x, \xi) = \sum_{|\mu| < N_1} \frac{(\partial_x^\mu \tau)(x, \xi)}{\mu!} A_k^\mu(x, \xi) + T_{N_1}^{(k)}(x, \xi), \quad (3.4.26)$$

where

$$A_k^\mu(x, \xi) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \xi} K_k(x, z) (-z)^\mu dz, \quad (3.4.27)$$

and the remainder term is

$$T_{N_1}^{(k)}(x, \xi) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \xi} K_k(x, z) R_{N_1}(x, z, \xi) dz. \quad (3.4.28)$$

Inserting the definition (3.4.10) of K_k into (3.4.27),

$$A_k^\mu(x, \xi) = (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{iz \cdot (\eta - \xi)} (-z)^\mu \sigma_k(x, \eta) d\eta dz.$$

The compact support of σ_k in η justifies both the interchange of integrals and the integrations by parts that follow. Observe the duality identity

$$\partial_\eta^\mu (e^{iz \cdot (\eta - \xi)}) = (iz)^\mu e^{iz \cdot (\eta - \xi)}, \quad (3.4.29)$$

from which $(-z)^\mu e^{iz \cdot (\eta - \xi)} = (-1)^{|\mu|} (-i)^{|\mu|} \partial_\eta^\mu (e^{iz \cdot (\eta - \xi)}) = i^{|\mu|} \partial_\eta^\mu (e^{iz \cdot (\eta - \xi)})$, using $(-1)^{|\mu|} (-i)^{|\mu|} = (-1)^{2|\mu|} i^{|\mu|} = i^{|\mu|}$. Hence

$$A_k^\mu(x, \xi) = i^{|\mu|} (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \partial_\eta^\mu (e^{iz \cdot (\eta - \xi)}) \sigma_k(x, \eta) d\eta dz.$$

Integration by parts in η (the boundary terms vanish since $\sigma_k(x, \cdot)$ has compact support) gives

$$\begin{aligned} A_k^\mu(x, \xi) &= i^{|\mu|} (-1)^{|\mu|} (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{iz \cdot (\eta - \xi)} (\partial_\eta^\mu \sigma_k)(x, \eta) d\eta dz \\ &= (-i)^{|\mu|} (2\pi)^{-n} \int_{\mathbb{R}^n} (\partial_\eta^\mu \sigma_k)(x, \eta) \left[\int_{\mathbb{R}^n} e^{iz \cdot (\eta - \xi)} dz \right] d\eta. \end{aligned}$$

By the Fourier inversion formula, $(2\pi)^{-n} \int e^{iz \cdot (\eta - \xi)} dz = \delta(\eta - \xi)$ in the sense of distributions, whence

$$A_k^\mu(x, \xi) = (-i)^{|\mu|} (\partial_\xi^\mu \sigma_k)(x, \xi). \quad (3.4.30)$$

Plugging (3.4.30) back into (3.4.26), we obtain

$$\lambda_k(x, \xi) = \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma_k)(x, \xi) (\partial_x^\mu \tau)(x, \xi) + T_{N_1}^{(k)}(x, \xi). \quad (3.4.31)$$

Set $\lambda(x, \xi) := \sum_{k=0}^{\infty} \lambda_k(x, \xi)$, so that T_λ is the symbol of $T_\sigma T_\tau$. Summing (3.4.31) over k and using $\sum_k \partial_\xi^\mu \sigma_k = \partial_\xi^\mu \sigma$,

$$\lambda(x, \xi) = \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(x, \xi) (\partial_x^\mu \tau)(x, \xi) + T_{N_1}(x, \xi), \text{ with } T_{N_1} = \sum_{k=0}^{\infty} T_{N_1}^{(k)}.$$

It remains to show that for every positive integer N and all multi-indices α, β , there exists $C_{\alpha, \beta} > 0$ such that

$$\left| \partial_x^\alpha \partial_\xi^\beta \left[\lambda - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \right] (x, \xi) \right| \leq C_{\alpha, \beta} (1 + |\xi|)^{m_1 + m_2 - N - |\beta|} \quad (3.4.32)$$

for all $(x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n$, provided N_1 is chosen sufficiently large. Since

$$\sum_{N \leq |\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \in S^{m_1 + m_2 - N},$$

it suffices, by (3.4.31) and the definition $T_{N_1} = \sum_k T_{N_1}^{(k)}$, to estimate each $T_{N_1}^{(k)}$ and sum over k . This is the content of Lemmas 3.4.2 and 3.4.1.

By the Lemma 3.4.2, for M large enough so that

$$(1 + |\xi|)^{m_2 - 2M} \leq (1 + |\xi|)^{m_1 + m_2 - N - |\beta|}, \quad (3.4.33)$$

This requires $2M \geq m_1 + N + |\beta|$, which can always be achieved by choosing M and N_1 large enough so that $m_1 + 2M - N_1 < 0$. The geometric series $\sum_{k=0}^{\infty} 2^{(m_1 + 2M - N_1)k}$ then converges, and

$$\left| \partial_x^\alpha \partial_\xi^\beta T_{N_1}(x, \xi) \right| \leq \sum_{k=0}^{\infty} C_{\alpha, \beta, M} (1 + |\xi|)^{m_2 - 2M} 2^{(m_1 + 2M - N_1)k} \leq C_{\alpha, \beta} (1 + |\xi|)^{m_1 + m_2 - N - |\beta|},$$

which establishes (3.4.32). Therefore $\lambda \in S^{m_1 + m_2}$ and admits the asymptotic expansion claimed in the theorem. $\square$

3.4.4 Remark. The factor $(1 + |\xi|)^{m_2 - 2M}$ controls the decay in ξ (made arbitrarily strong by choosing M large), while the factor $2^{(m_1 + 2M - N_1)k}$ ensures the convergence of the series $\sum_k T_{N_1}^{(k)}$ as soon as $N_1 > m_1 + 2M$, turning it into a convergent geometric series.

3.4.5 Theorem. (Wong, 2014) For all nonnegative integers N , and multi-indices α and β , there exists a constant A , depending on m, n, N, α and β only, such that

$$\int_{\mathbb{R}^n} |z|^N |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)| dz \leq A 2^{(m + |\alpha| - N)k}$$

for all $k = 0, 1, 2, \dots$

3.5 Formal adjoint of a pseudo-differential operator

For any pair of functions $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$ we write

$$(\varphi, \psi) = \int_{\mathbb{R}^n} \varphi(x) \overline{\psi(x)} dx. \quad (3.5.1)$$

Let $\sigma \in S^m$ and let T_σ be its associated pseudo-differential operator. A linear operator $T_\sigma^* : \mathcal{S} \rightarrow \mathcal{S}$ satisfying

$$(T_\sigma \varphi, \psi) = (\varphi, T_\sigma^* \psi), \quad \varphi, \psi \in \mathcal{S}(\mathbb{R}^n), \quad (3.5.2)$$

is called the *formal adjoint* of T_σ .

3.5.1 Theorem. *Let $\sigma \in S^m$. Then the formal adjoint of T_σ is again a pseudo-differential operator, $T_\sigma^* = T_\tau$, whose symbol τ belongs to S^m and admits the asymptotic expansion*

$$\tau(x, \xi) \sim \sum_{\mu} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\xi^\mu \overline{\sigma(x, \xi)}. \quad (3.5.3)$$

More precisely, for every positive integer N ,

$$\tau(x, \xi) - \sum_{|\mu| < N} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\xi^\mu \overline{\sigma(x, \xi)} \in S^{m-N}. \quad (3.5.4)$$

Proof. We first construct a candidate for τ explicitly. Let $\{\sigma_k\}_{k \geq 0}$ be the dyadic decomposition of σ in the frequency variable, so that $\sigma = \sum_{k \geq 0} \sigma_k$ with each σ_k compactly supported in ξ , and let

$$K_k(x, z) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \eta} \sigma_k(x, \eta) d\eta \quad (3.5.5)$$

be the partial kernel of T_{σ_k} , i.e. the inverse Fourier transform of σ_k in its second variable.

For $\varphi, \psi \in \mathcal{S}$, writing $T_{\sigma_k} \varphi$ through its kernel and applying Fubini's theorem (legitimate because σ_k has compact ξ -support, so $K_k(x, \cdot) \in \mathcal{S}$),

$$\begin{aligned} (T_{\sigma_k} \varphi, \psi) &= \int_{\mathbb{R}^n} (T_{\sigma_k} \varphi)(x) \overline{\psi(x)} dx = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} K_k(x, x-y) \varphi(y) dy \right\} \overline{\psi(x)} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} K_k(x, x-y) \overline{\psi(x)} dx \right\} \varphi(y) dy. \end{aligned} \quad (3.5.6)$$

On the other hand, by the definition (3.5.2) of the formal adjoint, $(T_{\sigma_k} \varphi, \psi) = (\varphi, T_{\sigma_k}^* \psi) = \int \varphi(y) \overline{(T_{\sigma_k}^* \psi)(y)} dy$. Comparing this with (3.5.6), valid for every $\varphi \in \mathcal{S}$, we obtain

$$(T_{\sigma_k}^* \psi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \overline{K_k(y, y-x)} \psi(y) dy. \quad (3.5.7)$$

Inserting the Fourier inversion formula for ψ , applying Fubini's theorem, and performing the change of variables $z = y - x$,

$$(T_{\sigma_k}^* \psi)(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} \overline{K_k(y, y-x)} \left\{ \int_{\mathbb{R}^n} e^{iy \cdot \eta} \widehat{\psi}(\eta) d\eta \right\} dy = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \eta} \tau_k(x, \eta) \widehat{\psi}(\eta) d\eta, \quad (3.5.8)$$

where

$$\tau_k(x, \eta) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} \overline{K_k(x+z, z)} dz. \quad (3.5.9)$$

Identity (3.5.8) shows that $T_{\sigma_k}^* = T_{\tau_k}$. Since $(T_\sigma \varphi, \psi) = \sum_{k \geq 0} (T_{\sigma_k} \varphi, \psi)$ for all $\varphi, \psi \in \mathcal{S}$, the natural candidate for the adjoint symbol is

$$\tau(x, \eta) = \sum_{k=0}^{\infty} \tau_k(x, \eta). \quad (3.5.10)$$

Fix a positive integer N_1 . For each $k \geq 0$, Taylor's formula with integral remainder applied to $z \mapsto K_k(x+z, z)$ around $z=0$ gives

$$K_k(x+z, z) = \sum_{|\mu| < N_1} \frac{z^\mu}{\mu!} (\partial_x^\mu K_k)(x, z) + R_{N_1}^{(k)}(x, z), \quad (3.5.11)$$

where

$$R_{N_1}^{(k)}(x, z) = N_1 \sum_{|\mu|=N_1} \frac{z^\mu}{\mu!} \int_0^1 (1-\theta)^{N_1-1} (\partial_x^\mu K_k)(x+\theta z, z) d\theta. \quad (3.5.12)$$

Substitute (3.5.11) into (3.5.9). For the polynomial terms, the identity $z^\mu e^{i\eta \cdot z} = (-iD_\eta)^\mu e^{i\eta \cdot z}$ followed by integration by parts in η (the boundary terms vanish since σ_k has compact ξ -support) yields

$$\tau_k(x, \eta) = \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\xi^\mu \overline{\sigma_k(x, \eta)} + T_{N_1}^{(k)}(x, \eta), \quad (3.5.13)$$

where the remainder symbol is

$$T_{N_1}^{(k)}(x, \eta) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} \overline{R_{N_1}^{(k)}(x, z)} dz. \quad (3.5.14)$$

Fix a positive integer N and choose $N_1 > N$. Summing (3.5.13) over k and using (3.5.10),

$$\tau - \sum_{|\mu| < N} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\xi^\mu \overline{\sigma} = \sum_{N \leq |\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\xi^\mu \overline{\sigma} + \sum_{k=0}^{\infty} T_{N_1}^{(k)}. \quad (3.5.15)$$

The first sum on the right-hand side lies in S^{m-N} : each term involves $N \leq |\mu| < N_1$ derivatives, and ∂_ξ^μ lowers the order of $\overline{\sigma} \in S^m$ by $|\mu| \geq N$.

For the second sum, fix multi-indices α, β . Integrating by parts in (3.5.14) by means of the operator $(1 - \Delta_z)^K$, using $(1 - \Delta_z)^K e^{i\eta \cdot z} = (1 + |\eta|^2)^K e^{i\eta \cdot z}$, gives

$$(D_x^\alpha D_\eta^\beta T_{N_1}^{(k)})(x, \eta) = (1 + |\eta|^2)^{-K} (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} (1 - \Delta_z)^K \left\{ z^\beta \overline{(D_x^\alpha R_{N_1}^{(k)})(x, z)} \right\} dz, \quad (3.5.16)$$

for any positive integer K . Applying Leibniz's rule and the explicit form (3.5.12) of the remainder, and using $\sigma \in S^m$ together with the standard kernel estimate for the dyadic pieces, one obtains: for every α, β there are integers K, M', N_1 , large enough (depending on α, β but not on k) with $m + 2K - 2M' + n < 0$, and a constant $C > 0$ independent of k , such that

$$|(D_x^\alpha D_\eta^\beta T_{N_1}^{(k)})(x, \eta)| \leq C (1 + |\eta|)^{m-N_1-|\beta|} 2^{k(m+2K-2M'+n)}. \quad (3.5.17)$$

Since the exponent $m+2K-2M'+n$ is negative, the geometric series $\sum_{k \geq 0} 2^{k(m+2K-2M'+n)}$ converges, and summing (3.5.17) over k we have a constant $C_{\alpha,\beta} > 0$ with

$$\left| D_x^\alpha D_\eta^\beta \sum_{k \geq 0} T_{N_1}^{(k)}(x, \eta) \right| \leq C_{\alpha,\beta} (1 + |\eta|)^{m-N_1-|\beta|}. \quad (3.5.18)$$

Since $N_1 > N$, the right-hand side is $O((1 + |\eta|)^{m-N-|\beta|})$, so $\sum_k T_{N_1}^{(k)} \in S^{m-N}$. Together with the first sum of (3.5.15), this proves (3.5.4).

Taking $N = 1$ in (3.5.4) gives $\tau - \bar{\sigma} \in S^{m-1}$, hence $\tau \in S^m$; in particular τ is a symbol and T_τ a pseudo-differential operator. The asymptotic expansion (3.5.3) is exactly the family of estimates (3.5.4) for all N . Finally, the identity $(T_\sigma \varphi, \psi) = (\varphi, T_\tau \psi)$ for all $\varphi, \psi \in \mathcal{S}$ follows by summing $T_{\sigma_k}^* = T_{\tau_k}$ over k , using (3.5.7)-(3.5.10), thus $T_\tau = T_\sigma^*$. $\square$

3.6 Ellipticity and parametrix

Among pseudo-differential operators, the elliptic ones occupy a special place: they possess approximate inverses, called *parametrix*s, which are again pseudo-differential operators.

3.6.1 Definition. (Wong, 2014) A symbol $\sigma \in S^m$ is called *elliptic* when one can find positive constants C and R for which

$$|\sigma(x, \xi)| \geq C (1 + |\xi|)^m, \quad |\xi| > R.$$

A pseudo-differential operator T_σ is said to be elliptic precisely when its symbol is elliptic.

3.6.2 Theorem. Let σ be an elliptic symbol in S^m . Then there is a symbol $\tau \in S^{-m}$ for which

$$T_\tau T_\sigma = I + R, \quad (3.6.1)$$

$$T_\sigma T_\tau = I + S, \quad (3.6.2)$$

where R and S are pseudo-differential operators whose symbols lie in $\bigcap_{k \in \mathbb{R}} S^k = S^{-\infty}$, and I denotes the identity operator.

Proof. Let us suppose that we have $\tau_j \in S^{-m-j}$, $j = 0, 1, 2, \dots$. Then, by the theorem of asymptotic expansion (Theorem 3.3.2), there exists $\tau \in S^{-m}$ such that $\tau \sim \sum_{j=0}^{\infty} \tau_j$. By the composition theorem (Theorem 3.4.3), the symbol λ associated to $T_\tau T_\sigma$ is given by

$$\lambda - \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau) (\partial_x^\gamma \sigma) \in S^{-N} \quad \text{for every positive integer } N.$$

We compute

$$\begin{aligned} \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau) (\partial_x^\gamma \sigma) &= \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} \left(\partial_\xi^\gamma \sum_{j=0}^{\infty} \tau_j \right) (\partial_x^\gamma \sigma) = \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} \left(\sum_{j=0}^{\infty} \partial_\xi^\gamma \tau_j \right) (\partial_x^\gamma \sigma) \\ &= \sum_{|\gamma| < N} \sum_{j=0}^{\infty} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) = \tau_0 \sigma + \sum_{\ell=1}^N \left[\tau_\ell \sigma + \sum_{\substack{|\gamma|+j=\ell \\ j < \ell}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \right] \\ &\quad + \sum_{\substack{|\gamma|+j > N \\ |\gamma| < N, j < N}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma). \end{aligned}$$

Let us take

$$\tau_0(x, \xi) = \begin{cases} \frac{\psi(\xi)}{\sigma(x, \xi)}, & |\xi| > r, \\ 0, & |\xi| \leq r, \end{cases} \quad r > 0,$$

where $\psi \in C^\infty(\mathbb{R}^n)$ is such that $\psi(\xi) = 1$ for $|\xi| \geq 2r$ and $\psi(\xi) = 0$ for $|\xi| \leq r$. So τ_0 is well defined since σ is elliptic.

In addition, define inductively, for every integer $\ell \geq 1$,

$$\tau_\ell = - \left\{ \sum_{\substack{|\gamma|+j=\ell \\ j < \ell}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \right\} \tau_0.$$

Then $\tau_j \in S^{-m-j}$ for every $j \geq 0$ (prove by induction on j). So $\tau_0 \sigma = 1$ for $|\xi| \geq 2r$, and the bracketed second term $\sum_{\ell=1}^N [\dots]$ in the previous expansion will become 0, together with the entire right side, when $|\xi| \leq r$ and when $|\xi| \geq 2r$.

Hence

$$\lambda - 1 = \sum_{\substack{|\gamma|+j > N \\ |\gamma| < N, j < N}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma),$$

but $\partial_\xi^\gamma \tau_j \in S^{-m-j-|\gamma|}$ and $\sigma \in S^m$ implies $\partial_x^\gamma \sigma \in S^m$, so $(\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \in S^{-(j+|\gamma|)}$. But $|\gamma| + j > N$ implies $-(j + |\gamma|) < -N$, then $\lambda - 1 \in S^{-N}$, $\forall N > 0$ (positive integer). Hence

$$\lambda - 1 \in \bigcap_{k \in \mathbb{R}} S^k.$$

Let us take R to be the pseudo-differential operator associated to $\lambda - 1$ (its symbol). So we have

$$T_\tau T_\sigma = I + R, \quad \text{with } R = T_{\lambda-1}.$$

With the same process, we can find $\kappa \in S^{-m}$ such that

$$T_\sigma T_\kappa = I + R',$$

with R' a pseudo-differential operator whose symbol lies in $\bigcap_{k \in \mathbb{R}} S^k$. Then

$$T_\tau T_\sigma T_\kappa = T_\kappa + R T_\kappa \quad \text{and} \quad T_\tau T_\sigma T_\kappa = T_\tau + T_\tau R',$$

so

$$T_\kappa + R T_\kappa = T_\tau + T_\tau R',$$

with $R T_\kappa$ and $T_\tau R'$ in $\bigcap_{k \in \mathbb{R}} S^k$. We obtain

$$T_\kappa = T_\tau + T_\tau R' - R T_\kappa = T_\tau + R'', \quad R'' = T_\tau R' - R T_\kappa,$$

where R'' has its symbol in $\bigcap_{k \in \mathbb{R}} S^k$. Therefore, multiplying $T_\kappa = T_\tau + R''$ on the left by T_σ ,

$$T_\sigma T_\kappa = T_\sigma T_\tau + T_\sigma R'' \implies I + R' = T_\sigma T_\tau + T_\sigma R'',$$

hence

$$T_\sigma T_\tau = I + S, \quad S = R' - T_\sigma R'',$$

and S is a pseudo-differential operator with symbol in $\bigcap_{k \in \mathbb{R}} S^k$. □

3.6.3 Definition. (The definition of Hörmander (2007)) Let $m \in \mathbb{R}$ and $0 \leq \delta, \rho \leq 1$. The symbol class $S_{\rho,\delta}^m(\mathbb{R}^n \times \mathbb{R}^n)$ consists of all functions

$$p(x, \xi) \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$$

such that, for every pair of multi-indices $\alpha, \beta \in \mathbb{N}^n$, there exists a constant $C_{\alpha,\beta} > 0$ satisfying

$$\left| D_x^\alpha D_\xi^\beta p(x, \xi) \right| \leq C_{\alpha,\beta} (1 + |\xi|)^{m - \rho|\beta| + \delta|\alpha|}, \quad \forall (x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n.$$

3.6.4 Remark. The results of this chapter were stated for the standard class $S^m = S_{1,0}^m$. They extend to the Hörmander classes $S_{\rho,\delta}^m$ ($0 < \rho \leq 1$, $0 \leq \delta < 1$), defined by 3.6.3, as follows (Hörmander, 2007, §18.1, p. 94). The asymptotic summation (Theorem 3.3.2) holds without change, and the L^2 - and Sobolev-continuity hold for $\delta \leq \rho$. The composition (Theorem 3.4.3), the adjoint (Theorem 3.5.1) and the elliptic parametrix (Theorem 3.6.2) remain valid *provided* $\delta < \rho$: the α -th term of the composition expansion (3.4.19) lies in $S_{\rho,\delta}^{m_1+m_2-(\rho-\delta)|\alpha|}$, so its order decreases only when $\rho - \delta > 0$; at $\delta = \rho$ the asymptotic calculus collapses. This borderline is exactly the obstruction met in this chapter.

4. Parametrix and adjoint formulation for fractional Bessel potential with drift in supercritical regime.

In this chapter, we study the operator $\mathcal{L} = (I - \Delta)^s + \nu(x) \cdot \nabla$ in the supercritical regime $s \in (0, 1/2)$ with $\nu \neq 0$. After establishing the symbolic structure of $\mathcal{L}$ and the failure of global ellipticity, we exhibit a directional ellipticity property and construct a two-sided parametrix for $\mathcal{L}$ in the Hörmander class $S_{2s, 1-2s}^{-2s}$ in the sub-regime $s \in (1/4, 1/2)$. The formal L^2 -adjoint is then computed explicitly. The remaining range $s \in (0, 1/4]$ requires more elaborate techniques, which we discuss as perspectives at the end of the chapter.

4.1 The fractional Bessel potential with drift ($0 < s < 1/2$).

4.1.1 Theorem. *Let $s \in (0, 1/2)$, and let $\nu = (\nu_1, \dots, \nu_n) : \mathbb{R}^n \rightarrow \mathbb{R}^n$, be a smooth vector field all of whose partial derivatives are bounded on $\mathbb{R}^n$. Consider the operator*

$$\mathcal{L} = (I - \Delta)^s + \nu(x) \cdot \nabla, \quad \nu(x) \cdot \nabla = \sum_{j=1}^n \nu_j(x) \partial_j.$$

Then $\mathcal{L}$ is a pseudo-differential operator whose symbol

$$\sigma(x, \xi) = (1 + |\xi|^2)^s + i \nu(x) \cdot \xi$$

belongs to S^1 .

Proof. We first recall how differentiation behaves under the Fourier transform. For $u \in \mathcal{S}(\mathbb{R}^n)$ and $j \in \{1, \dots, n\}$, integration by parts gives

$$\widehat{\partial_j u}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} (\partial_j u)(x) dx = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (i\xi_j) e^{-ix \cdot \xi} u(x) dx = (i\xi_j) \widehat{u}(\xi).$$

Iterating once more,

$$\widehat{\partial_j^2 u}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (i\xi_j)^2 e^{-ix \cdot \xi} u(x) dx = -\xi_j^2 \widehat{u}(\xi),$$

and therefore, summing over j with $\Delta = \sum_{j=1}^n \partial_j^2$,

$$\widehat{\Delta u}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \sum_{j=1}^n (i\xi_j)^2 e^{-ix \cdot \xi} u(x) dx = -|\xi|^2 \widehat{u}(\xi).$$

Hence

$$-\widehat{\Delta u}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} |\xi|^2 e^{-ix \cdot \xi} u(x) dx = |\xi|^2 \widehat{u}(\xi).$$

In fact, we have

$$((I - \Delta)u)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (1 + |\xi|^2) e^{ix \cdot \xi} \widehat{u}(\xi) d\xi,$$

so that

$$(\widehat{(I - \Delta)u})(\xi) = (1 + |\xi|^2) \widehat{u}(\xi).$$

From the spectral theorem for multiplication [A.1.1](#) ([Haase, 2006](#)) and the inverse Fourier transform we get

$$((I - \Delta)^s u)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} (1 + |\xi|^2)^s \hat{u}(\xi) d\xi. \quad (4.1.1)$$

We now treat the first-order term. For $u \in \mathcal{S}(\mathbb{R}^n)$, using $\partial_j u(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (i\xi_j) e^{ix \cdot \xi} \hat{u}(\xi) d\xi$, we have

$$\begin{aligned} (\nu(x) \cdot \nabla) u(x) &= \sum_{j=1}^n \nu_j(x) (\partial_j u)(x) = \sum_{j=1}^n \nu_j(x) (2\pi)^{-n/2} \int_{\mathbb{R}^n} (i\xi_j) e^{ix \cdot \xi} \hat{u}(\xi) d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \left(\sum_{j=1}^n i \nu_j(x) \xi_j \right) \hat{u}(\xi) d\xi = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} (i \nu(x) \cdot \xi) \hat{u}(\xi) d\xi. \end{aligned}$$

Adding this to [\(4.1.1\)](#), we obtain

$$(\mathcal{L}u)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \left[(1 + |\xi|^2)^s + i \nu(x) \cdot \xi \right] \hat{u}(\xi) d\xi.$$

Comparing with Definition [3.1.2](#), we see that $\mathcal{L}$ is the pseudo-differential operator $\mathcal{L}$ with

$$\sigma(x, \xi) = (1 + |\xi|^2)^s + i \nu(x) \cdot \xi.$$

It remains to show that $\sigma \in S^1$. We estimate the two summands separately.

The term $(1 + |\xi|^2)^s$. We claim that for every multi-index α there is a positive constant $C_{n,\alpha}$ such that

$$|D_\xi^\alpha (1 + |\xi|^2)^s| \leq C_{n,\alpha} (1 + |\xi|)^{2s - |\alpha|}, \quad \xi \in \mathbb{R}^n. \quad (4.1.2)$$

We argue by induction on the length $|\alpha|$, the estimate being required for every $s \in \mathbb{R}$.

For $\alpha = 0$, since $|\xi|^2 \leq (1 + |\xi|)^2$,

$$|(1 + |\xi|^2)^s| \leq C (1 + |\xi|)^{2s},$$

which is [\(4.1.2\)](#) with $|\alpha| = 0$.

Assume [\(4.1.2\)](#) holds for every multi-index of length ℓ (and every $s \in \mathbb{R}$). Let α' have length $\ell + 1$. Then $D_\xi^{\alpha'} = D_\xi^\alpha \partial_{\xi_j}$ for some α of length ℓ and some $j \in \{1, \dots, n\}$, and

$$D_\xi^{\alpha'} (1 + |\xi|^2)^s = D_\xi^\alpha [2s \xi_j (1 + |\xi|^2)^{s-1}] = 2s \sum_{\delta \leq \alpha} \binom{\alpha}{\delta} (D_\xi^\delta \xi_j) D_\xi^{\alpha-\delta} (1 + |\xi|^2)^{s-1}.$$

Here $D_\xi^\delta \xi_j$ vanishes unless $|\delta| \leq 1$, and in all cases $|D_\xi^\delta \xi_j| \leq 1 + |\xi|$, since for $\delta = 0$ one has $|\xi_j| \leq |\xi| \leq 1 + |\xi|$ and for $|\delta| = 1$ one has $|D_\xi^\delta \xi_j| \in \{0, 1\} \leq 1 + |\xi|$. Applying the induction hypothesis to $D_\xi^{\alpha-\delta} (1 + |\xi|^2)^{s-1}$ (length $\leq \ell$, exponent $s - 1$),

$$\begin{aligned} |D_\xi^{\alpha'} (1 + |\xi|^2)^s| &\leq 2|s| \sum_{\delta \leq \alpha} \binom{\alpha}{\delta} (1 + |\xi|)^{1-|\delta|} C_{n,\alpha-\delta} (1 + |\xi|)^{2(s-1)-|\alpha|+|\delta|} \\ &= 2|s| \sum_{\delta \leq \alpha} \binom{\alpha}{\delta} C_{n,\alpha-\delta} (1 + |\xi|)^{2s-1-|\alpha|} = C_{n,\alpha'} (1 + |\xi|)^{2s-|\alpha'|}, \end{aligned}$$

with $C_{n,\alpha'} = 2|s| \sum_{\delta \leq \alpha} \binom{\alpha}{\delta} C_{n,\alpha-\delta}$ and $|\alpha'| = |\alpha| + 1$. This proves (4.1.2), so $(1 + |\xi|^2)^s \in S^{2s}$. Since $s \leq \frac{1}{2}$, we have $2s \leq 1$, and therefore $(1 + |\xi|)^{2s-|\alpha|} \leq (1 + |\xi|)^{1-|\alpha|}$, which shows $(1 + |\xi|^2)^s \in S^1$.

Write $\sigma_1(x, \xi) = i \sum_{j=1}^n \nu_j(x) \xi_j$, for multi-indices α, β ,

$$D_\xi^\alpha D_x^\beta \sigma_1(x, \xi) = i \sum_{j=1}^n (D_x^\beta \nu_j(x)) (D_\xi^\alpha \xi_j).$$

The factor $D_\xi^\alpha \xi_j$ vanishes for $|\alpha| \geq 2$ and equals 0 or 1 for $|\alpha| \leq 1$; moreover each $D_x^\beta \nu_j$ is bounded on $\mathbb{R}^n$ by hypothesis. Hence there is a constant $C_{\alpha,\beta} > 0$ such that

$$|D_\xi^\alpha D_x^\beta \sigma_1(x, \xi)| \leq C_{\alpha,\beta} (1 + |\xi|)^{1-|\alpha|}, \quad x, \xi \in \mathbb{R}^n,$$

since for $\alpha = 0$ the right-hand side is $C|\xi| \leq C(1 + |\xi|)$, for $|\alpha| = 1$ it is a bounded quantity $\leq C(1 + |\xi|)^0$, and for $|\alpha| \geq 2$ it is zero. Therefore $\sigma_1 \in S^1$.

Since $\sigma = (1 + |\xi|^2)^s + \sigma_1$ with both summands in S^1 , we conclude $\sigma \in S^1$. Therefore $\mathcal{L} = (I - \Delta)^s + \nu(x) \cdot \nabla$ is a pseudo-differential operator of order 1. $\square$

4.1.2 Proposition. Let $s \in (0, \frac{1}{2})$ and let $\nu \in C_b^\infty(\mathbb{R}^n; \mathbb{R}^n)$ ($n \geq 2$), i.e., ν is a smooth vector field all of whose partial derivatives are bounded on $\mathbb{R}^n$. Then the symbol

$$\sigma(x, \xi) = (1 + |\xi|^2)^s + i \nu(x) \cdot \xi$$

has order $m = 1$, that is $\sigma \in S^1$, but σ is *not* elliptic.

Proof. That $\sigma \in S^1$ has already been established in Theorem 4.1.1: the term $(1 + |\xi|^2)^s$ lies in S^{2s} and, since $2s < 1$, also in S^1 , while the term $i \nu(x) \cdot \xi$ lies in S^1 because ν and all its derivatives are bounded. Hence $\sigma \in S^1$, and we now check that the order cannot be lowered.

We assume $\nu \neq 0$, so that there is a point x_0 with $\nu(x_0) \neq 0$. Choose ξ parallel to $\nu(x_0)$, say $\xi = t \nu(x_0)$ with $t > 0$. Then

$$|\sigma(x_0, \xi)| \geq |\operatorname{Im} \sigma(x_0, \xi)| = |\nu(x_0) \cdot \xi| = t |\nu(x_0)|^2 = |\nu(x_0)| |\xi|.$$

Thus $|\sigma(x_0, \xi)|$ grows at least linearly in $|\xi|$, so σ cannot belong to $S^{m'}$ for any $m' < 1$; combined with $\sigma \in S^1$, the order of σ is exactly $m = 1$.

Let us now verify if the symbol σ is actually not elliptic. We argue by contradiction. Suppose σ were elliptic. Then, by Definition 3.6.1, there would exist constants $C > 0$ and $R > 0$ such that

$$|\sigma(x, \xi)| \geq C(1 + |\xi|)^1, \quad x \in \mathbb{R}^n, \quad |\xi| \geq R. \quad (4.1.3)$$

Fix any point $x \in \mathbb{R}^n$ and choose ξ *orthogonal* to $\nu(x)$, that is $\nu(x) \cdot \xi = 0$. Such a ξ exists with $|\xi|$ arbitrarily large: if $\nu(x) = 0$ every ξ works, and if $\nu(x) \neq 0$ the hyperplane $\{\xi : \nu(x) \cdot \xi = 0\}$ is $(n-1)$ -dimensional and unbounded (here we use $n \geq 2$). For such a ξ , the imaginary part of σ vanishes and $\sigma(x, \xi) = (1 + |\xi|^2)^s$, hence $|\sigma(x, \xi)| = (1 + |\xi|^2)^s \leq ((1 + |\xi|)^2)^s = (1 + |\xi|)^{2s}$. Combining this with the ellipticity inequality (4.1.3), which must hold for this ξ as soon as $|\xi| \geq R$, we obtain $C(1 + |\xi|) \leq |\sigma(x, \xi)| \leq (1 + |\xi|)^{2s}$, that is, $C \leq (1 + |\xi|)^{2s-1}$, $|\xi| \geq R$, $\nu(x) \cdot \xi = 0$. Since $s < \frac{1}{2}$, we have $2s - 1 < 0$, so $(1 + |\xi|)^{2s-1} \rightarrow 0$ as $|\xi| \rightarrow \infty$. Letting $|\xi| \rightarrow \infty$ along the hyperplane $\{\nu(x) \cdot \xi = 0\}$ forces $C \leq 0$, which contradicts $C > 0$.

Therefore no such constants C, R exist, and σ is not elliptic. $\square$

4.1.3 Lemma (Directional ellipticity). Let $s \in (0, 1/2)$ and $\nu \in C_b^\infty(\mathbb{R}^n; \mathbb{R}^n)$, and let $x_0 \in \mathbb{R}^n$ be such that $\nu(x_0) \neq 0$. For $x \in V$ (a neighbourhood of x_0) let

$$\Sigma_x = \{\xi \in \mathbb{R}^n : \nu(x) \cdot \xi = 0\}$$

denote the characteristic hyperplane at x , and for $\theta \in (0, 1]$ let

$$\Gamma_{x,\theta} = \{\xi \in \mathbb{R}^n \setminus \{0\} : |\nu(x) \cdot \xi| \geq \theta |\nu(x)| |\xi|\}$$

be the closed cone of aperture θ around the direction of $\nu(x)$. Then for every $\theta \in (0, 1]$ there exist a constant $C = C(\theta) > 0$ and a neighbourhood V of x_0 such that

$$|\sigma(x, \xi)| \geq C(1 + |\xi|), \quad x \in V, \quad |\xi| \geq 1, \quad \xi \in \Gamma_{x,\theta}. \quad (4.1.4)$$

In contrast, on the characteristic set one has

$$\sigma(x, \xi) = \langle \xi \rangle^{2s} \quad \text{whenever } \xi \in \Sigma_x,$$

so that $|\sigma(x, \xi)| = \langle \xi \rangle^{2s}$ is only of order $2s < 1$ there.

The ellipticity of σ thus degenerates completely in the characteristic directions: this is an anisotropic loss of ellipticity.

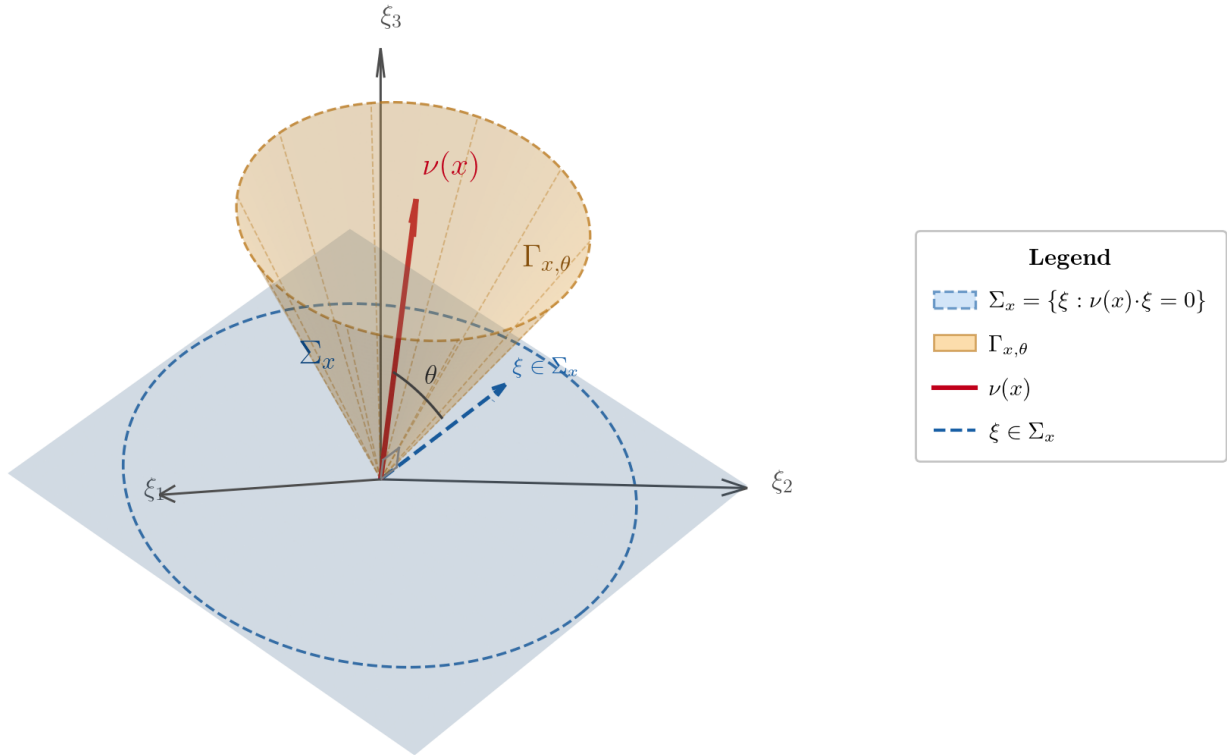


Figure 4.1: Geometry associated with the drift direction $\nu(x)$. The cone $\Gamma_{x,\theta}$ collects frequencies forming an angle smaller than θ with $\nu(x)$, while $\Sigma_x = \{\xi : \nu(x) \cdot \xi = 0\}$ represents the characteristic hyperplane.

Proof. Since ν is continuous and $\nu(x_0) \neq 0$, there is a neighbourhood V of x_0 on which

$$c_0 := \inf_{x \in V} |\nu(x)| > 0.$$

The symbol splits into its real and imaginary parts,

$$\operatorname{Re} \sigma(x, \xi) = (1 + |\xi|^2)^s, \quad \operatorname{Im} \sigma(x, \xi) = \nu(x) \cdot \xi,$$

so that

$$|\sigma(x, \xi)| \geq |\operatorname{Im} \sigma(x, \xi)| = |\nu(x) \cdot \xi|.$$

Let $\xi \in \Gamma_{x, \theta}$ with $|\xi| \geq 1$ and $x \in V$. By the definition of the cone and the bound $|\nu(x)| \geq c_0$,

$$|\sigma(x, \xi)| \geq |\nu(x) \cdot \xi| \geq \theta |\nu(x)| |\xi| \geq \theta c_0 |\xi|.$$

Finally, for $|\xi| \geq 1$ one has $1 + |\xi| \leq 2|\xi|$, hence $|\xi| \geq \frac{1}{2}(1 + |\xi|)$, and therefore

$$|\sigma(x, \xi)| \geq \theta c_0 |\xi| \geq \frac{\theta c_0}{2} (1 + |\xi|).$$

This is (4.1.4) with $C = C(\theta) = \theta c_0/2$.

If $\xi \in \Sigma_x$, then $\nu(x) \cdot \xi = 0$, so the imaginary part of σ vanishes and

$$\sigma(x, \xi) = (1 + |\xi|^2)^s = \langle \xi \rangle^{2s}.$$

Hence $|\sigma(x, \xi)| = \langle \xi \rangle^{2s}$ on Σ_x . Since $2s < 1$, the quantity $\langle \xi \rangle^{2s}$ cannot be bounded below by any multiple of $(1 + |\xi|)$ as $|\xi| \rightarrow \infty$. The lower bound (4.1.4) therefore requires the cone condition $\xi \in \Gamma_{x, \theta}$, and it deteriorates as $\theta \rightarrow 0$, the constant $C(\theta) = \theta c_0/2$ tending to 0. $\square$

4.1.4 Remark. The hypothesis “ $\xi \notin \Sigma_x$ ” alone would *not* yield (4.1.4): it only gives $\nu(x) \cdot \xi \neq 0$, hence $|\nu(x) \cdot \xi| > 0$, but this quantity may be arbitrarily small for ξ large and nearly orthogonal to $\nu(x)$. A *uniform* lower bound of order 1 requires ξ to stay in a closed cone $\Gamma_{x, \theta}$ strictly separated from the characteristic hyperplane Σ_x . The symbol σ is thus elliptic of order 1 only *microlocally*, on the conic region $\bigcup_x \{x\} \times \Gamma_{x, \theta}$, and completely non-elliptic along the characteristic directions Σ_x , where it drops to order $2s < 1$.

The symbol $\sigma(x, \xi) = (1 + |\xi|^2)^s + i \nu(x) \cdot \xi$ never vanishes, since $|\sigma(x, \xi)| \geq \operatorname{Re} \sigma(x, \xi) = (1 + |\xi|^2)^s \geq 1$; and, because $\sigma \in S^1$ with coefficients in C_b^∞ , the operator $\mathcal{L}$ maps $\mathcal{S}(\mathbb{R}^n)$ continuously into itself. Yet non-vanishing is *strictly weaker than ellipticity*. As Proposition 4.1.2 shows, σ is *not* elliptic. This is precisely why $\mathcal{L}$ need not admit a parametrix in the standard calculus, and it motivates the alternative approach developed below.

4.2 Adjoint construction for $s \in (0, 1/2)$

4.2.1 Proposition. For $\varphi \in C_c^\infty(\mathbb{R}^n)$, the formal L^2 adjoint $\mathcal{L}^*$ of $\mathcal{L}$ has symbol is

$$\sigma^*(x, \xi) = \langle \xi \rangle^{2s} - i \nu(x) \cdot \xi - \operatorname{div} \nu(x) + u(x, \xi),$$

where $u \in S_{1,0}^{-1}(\mathbb{R}^n)$.

In particular, the principal symbol of $\mathcal{L}^*$ is $\langle \xi \rangle^{2s} - i \nu(x) \cdot \xi$ (Ruzhansky, 2014).

Proof. Let $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$.

By the definition of the L^2 inner product and the oscillatory-kernel representation of $\mathcal{L}$,

$$\langle \mathcal{L}\varphi, \psi \rangle = \int_{\mathbb{R}^n} (\mathcal{L}\varphi)(x) \overline{\psi(x)} dx = (2\pi)^{-n} \int_{\mathbb{R}^n} \left[\iint_{\mathbb{R}^{2n}} e^{i(x-y)\cdot\xi} \sigma(x, \xi) \varphi(y) dy d\xi \right] \overline{\psi(x)} dx.$$

Since $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$ and $\sigma \in S_{1,0}^1$ (since $\nu(x) \cdot \xi$ has order 1 and $2s < 1$), the triple integral is absolutely convergent. Applying Fubini's theorem to interchange the order of integration yields

$$\langle \mathcal{L}\varphi, \psi \rangle = (2\pi)^{-n} \iiint_{\mathbb{R}^{3n}} e^{i(x-y)\cdot\xi} \sigma(x, \xi) \varphi(y) \overline{\psi(x)} dy dx d\xi. \quad (4.2.1)$$

We rewrite (4.2.1) in the form $\langle \varphi, \mathcal{L}^*\psi \rangle = \int_{\mathbb{R}^n} \varphi(y) \overline{(\mathcal{L}^*\psi)(y)} dy$. Using $e^{i(x-y)\cdot\xi} = \overline{e^{i(y-x)\cdot\xi}}$, the integrand takes the form $\varphi(y) \cdot (\dots)$, so integrating first over (x, ξ) gives

$$\langle \mathcal{L}\varphi, \psi \rangle = \int_{\mathbb{R}^n} \varphi(y) \underbrace{(2\pi)^{-n} \iint_{\mathbb{R}^{2n}} e^{i(y-x)\cdot\xi} \overline{\sigma(x, \xi)} \psi(x) dx d\xi}_{=:(\mathcal{L}^*\psi)(y)} dy.$$

Hence

$$(\mathcal{L}^*\psi)(y) = (2\pi)^{-n} \iint_{\mathbb{R}^{2n}} e^{i(y-x)\cdot\xi} \overline{\sigma(x, \xi)} \psi(x) dx d\xi. \quad (4.2.2)$$

This expression is not yet in standard PsDO form because $\overline{\sigma(x, \xi)}$ depends on the integration variable x rather than on the output variable y .

We apply Theorem 2.6.2 (Taylor's formula with integral remainder) to $f = \overline{\sigma(\cdot, \xi)} \in C^\infty(\mathbb{R}^n)$, expanding around the point y with increment $\eta = x - y$, for a fixed integer $N \geq 2$:

$$\overline{\sigma(x, \xi)} = \sum_{|\alpha| < N} \frac{(x-y)^\alpha}{\alpha!} \partial_y^\alpha \overline{\sigma(y, \xi)} + N \sum_{|\gamma|=N} \frac{(x-y)^\gamma}{\gamma!} \int_0^1 (1-\theta)^{N-1} (\partial_y^\gamma \overline{\sigma})(y + \theta(x-y), \xi) d\theta. \quad (4.2.3)$$

We recall the key identity, obtained by direct differentiation of the exponential in ξ :

$$(x-y)^\alpha e^{i(y-x)\cdot\xi} = (i\partial_\xi)^\alpha e^{i(y-x)\cdot\xi}. \quad (4.2.4)$$

Substituting (4.2.3) into (4.2.2) and using (4.2.4), we split $(\mathcal{L}^*\psi)(y)$ into a *principal part* and a *remainder*:

$$(\mathcal{L}^*\psi)(y) = P_N(y) + R_N(y), \quad (4.2.5)$$

where

$$P_N(y) = (2\pi)^{-n} \sum_{|\alpha| < N} \frac{1}{\alpha!} \iint_{\mathbb{R}^{2n}} (i\partial_\xi)^\alpha \left[e^{i(y-x)\cdot\xi} \right] \partial_y^\alpha \overline{\sigma(y, \xi)} \psi(x) dx d\xi, \quad (4.2.6)$$

$$R_N(y) = (2\pi)^{-n} N \sum_{|\gamma|=N} \frac{1}{\gamma!} \int_0^1 (1-\theta)^{N-1} \iint_{\mathbb{R}^{2n}} (i\partial_\xi)^\gamma \left[e^{i(y-x)\cdot\xi} \right] (\partial_y^\gamma \overline{\sigma})(y + \theta(x-y), \xi) \psi(x) dx d\xi d\theta. \quad (4.2.7)$$

In $P_N(y)$ we transfer the $|\alpha|$ derivatives in ξ onto the symbol by integration by parts; boundary terms vanish by Schwartz-class decay. Using $(-i)^{|\alpha|} \cdot i^{|\alpha|} = 1$:

$$P_N(y) = (2\pi)^{-n} \sum_{|\alpha| < N} \frac{(-i)^{|\alpha|}}{\alpha!} \iint_{\mathbb{R}^{2n}} e^{i(y-x)\cdot\xi} \partial_\xi^\alpha \partial_y^\alpha \overline{\sigma(y, \xi)} \psi(x) dx d\xi. \quad (4.2.8)$$

The same integration by parts applies to $R_N(y)$. Since $\sigma \in S_{1,0}^1$ (the term $i\nu(x) \cdot \xi$ has order 1, which dominates $\langle \xi \rangle^{2s}$ as $2s < 1$), every derivative $\partial_y^\gamma \sigma \in S_{1,0}^1$ for $|\gamma| = N$, and after the integration by parts in ξ the remainder lies in $S_{1,0}^{1-N}$. For $N \geq 2$ gives $1 - N \leq -1 < 0$, so

$$R_N \in S_{1,0}^{1-N}(\mathbb{R}^n) \subseteq S_{1,0}^{-1}(\mathbb{R}^n). \quad (4.2.9)$$

Combining (4.2.8) and (4.2.9) with $N \geq 2$, we obtain the PsDO representation:

$$(\mathcal{L}^* \psi)(y) = (2\pi)^{-n} \iint_{\mathbb{R}^{2n}} e^{i(y-x) \cdot \xi} \sigma^*(y, \xi) \psi(x) dx d\xi, \quad (4.2.10)$$

with adjoint symbol

$$\sigma^*(y, \xi) = \sum_{|\alpha| < 2} \frac{(-i)^{|\alpha|}}{\alpha!} \partial_\xi^\alpha \partial_y^\alpha \overline{\sigma(y, \xi)} + u(x, \xi), \quad u \in S_{1,0}^{-1}(\mathbb{R}^n). \quad (4.2.11)$$

If $|\alpha| = 0$.

$$\frac{i^0}{0!} \overline{\sigma(y, \xi)} = \langle \xi \rangle^{2s} - i \nu(y) \cdot \xi.$$

If $|\alpha| = 1$. For each unit multi-index $\alpha = e_j$, $j = 1, \dots, n$, with factor $(-i)^1/1! = -i$:

$$\partial_{\xi_j} \partial_{y_j} \overline{\sigma(y, \xi)} = \partial_{\xi_j} \partial_{y_j} (\langle \xi \rangle^{2s} - i \nu(y) \cdot \xi) = -i \sum_{k=1}^n \partial_{\xi_j} \partial_{y_j} (\nu_k(y) \cdot \xi_k) = -i \partial_{y_j} \nu_j(y) \quad (4.2.12)$$

Summing over j :

$$\sum_{j=1}^n -i \cdot (-i \partial_{y_j} \nu_j(y)) = -\operatorname{div} \nu(y).$$

The terms $|\alpha| < 2$ and renaming $y \mapsto x$:

$$\sigma^*(x, \xi) = \langle \xi \rangle^{2s} - i \nu(x) \cdot \xi - \operatorname{div} \nu(x) + u(x, \xi), \quad u \in S_{1,0}^{-1}(\mathbb{R}^n).$$

□

4.3 Parametrix construction for $(1/4 < s < 1/2)$.

4.3.1 Lemma. Let $s \in (0, \frac{1}{2})$ and let $\nu \in C_b^\infty(\mathbb{R}^n; \mathbb{R}^n)$. The symbol

$$\lambda(x, \xi) := \frac{1}{\sigma(x, \xi)} = \frac{1}{\langle \xi \rangle^{2s} + i \nu(x) \cdot \xi},$$

where $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$, is well defined and satisfies, for all multi-indices $\alpha, \beta \in \mathbb{N}^n$,

$$|D_\xi^\alpha D_x^\beta \lambda(x, \xi)| \lesssim \langle \xi \rangle^{-2s-2s|\alpha|+(1-2s)|\beta|}.$$

In particular $\lambda \in S_{2s, 1-2s}^{-2s}(\mathbb{R}^n \times \mathbb{R}^n)$.

Proof. We compute $D_\xi^\alpha D_x^\beta \lambda$ for $\lambda(x, \xi) = \frac{1}{\sigma(x, \xi)}$, where $\sigma(x, \xi) = \langle \xi \rangle^{2s} + i \nu(x) \cdot \xi$.

$$D_{x_j} \left(\frac{1}{\sigma} \right) = - \frac{D_{x_j} \sigma}{\sigma^2}.$$

For a general multi-index β , iterating and grouping terms gives

$$D_x^\beta \lambda = D_x^\beta \left(\frac{1}{\sigma} \right) = \sum_{k=1}^{|\beta|} \frac{(-1)^k k!}{\sigma^{k+1}} \sum_{\substack{\beta_1 + \dots + \beta_k = \beta \\ |\beta_i| \geq 1}} c_{\beta_1, \dots, \beta_k} \prod_{i=1}^k (D_x^{\beta_i} \sigma). \quad (4.3.1)$$

Now, since $\langle \xi \rangle^{2s}$ does not depend on x ,

$$D_x^{\beta_i} \sigma(x, \xi) = D_x^{\beta_i} (\langle \xi \rangle^{2s} + i \nu(x) \cdot \xi) = i (D_x^{\beta_i} \nu(x)) \cdot \xi.$$

Substituting this into (4.3.1),

$$D_x^\beta \lambda = \sum_{k=1}^{|\beta|} \frac{(-1)^k k! i^k}{\sigma^{k+1}} \sum_{\substack{\beta_1 + \dots + \beta_k = \beta \\ |\beta_i| \geq 1}} c_{\beta_1, \dots, \beta_k} \prod_{i=1}^k ((D_x^{\beta_i} \nu(x)) \cdot \xi). \quad (4.3.2)$$

We must differentiate (4.3.2) in ξ . Each term of the sum is a product of two ξ -dependent factors:

$$\underbrace{\frac{1}{\sigma^{k+1}}}_{\text{factor } A} \quad \text{and} \quad \underbrace{\prod_{i=1}^k ((D_x^{\beta_i} \nu(x)) \cdot \xi)}_{\text{factor } B}.$$

By the Leibniz rule, for a fixed term,

$$D_\xi^\alpha \left(\frac{1}{\sigma^{k+1}} \cdot \prod_{i=1}^k ((D_x^{\beta_i} \nu) \cdot \xi) \right) = \sum_{\alpha' \leq \alpha} \binom{\alpha}{\alpha'} (D_\xi^{\alpha'} \prod_{i=1}^k ((D_x^{\beta_i} \nu) \cdot \xi)) \left(D_\xi^{\alpha - \alpha'} \frac{1}{\sigma^{k+1}} \right). \quad (4.3.3)$$

Computation of factor B: the term $D_\xi^{\alpha'} \prod_{i=1}^k ((D_x^{\beta_i} \nu) \cdot \xi)$. Each factor $(D_x^{\beta_i} \nu(x)) \cdot \xi = \sum_{\ell=1}^n (D_x^{\beta_i} \nu_\ell)(x) \xi_\ell$ is linear in ξ . Hence its first ξ -derivative is constant in ξ and its second ξ -derivative vanishes:

$$D_\xi^\gamma ((D_x^{\beta_i} \nu) \cdot \xi) = \begin{cases} (D_x^{\beta_i} \nu) \cdot \xi, & \gamma = 0, \\ D_x^{\beta_i} \nu_\ell, & \gamma = e_\ell \ (|\gamma| = 1), \\ 0, & |\gamma| \geq 2. \end{cases}$$

Applying the Leibniz rule once more to the product of the k linear factors, the multi-index α' is split as $\alpha' = \gamma_1 + \dots + \gamma_k$, and only the splittings with $|\gamma_i| \leq 1$ for every i survive:

$$D_\xi^{\alpha'} \prod_{i=1}^k ((D_x^{\beta_i} \nu) \cdot \xi) = \sum_{\substack{\gamma_1 + \dots + \gamma_k = \alpha' \\ |\gamma_i| \leq 1}} \binom{\alpha'}{\gamma_1, \dots, \gamma_k} \prod_{i=1}^k D_\xi^{\gamma_i} ((D_x^{\beta_i} \nu) \cdot \xi).$$

This sum is empty (hence the whole term vanishes) as soon as $|\alpha'| > k$, since then at least one γ_i would have $|\gamma_i| \geq 2$. So in B only $|\alpha'| \leq k$ contributes.

Computation of factor A: the term $D_\xi^{\alpha-\alpha'} \frac{1}{\sigma^{k+1}}$. We again use the Faà di Bruno formula, now in ξ , for the function $\xi \mapsto \sigma^{-(k+1)}$. Writing $\mu := \alpha - \alpha'$,

$$D_\xi^\mu \frac{1}{\sigma^{k+1}} = \sum_{r=1}^{|\mu|} \frac{(-1)^r (k+r)!/k!}{\sigma^{k+1+r}} \sum_{\substack{\mu_1+\dots+\mu_r=\mu \\ |\mu_p| \geq 1}} c_{\mu_1, \dots, \mu_r} \prod_{p=1}^r (D_\xi^{\mu_p} \sigma),$$

and, since $D_\xi^{\mu_p} \sigma = D_\xi^{\mu_p} \langle \xi \rangle^{2s} + i \nu \cdot D_\xi^{\mu_p} \xi$,

$$D_\xi^{\mu_p} \sigma = \begin{cases} D_\xi^{\mu_p} \langle \xi \rangle^{2s} + i \nu_\ell, & |\mu_p| = 1, \\ D_\xi^{\mu_p} \langle \xi \rangle^{2s}, & |\mu_p| \geq 2. \end{cases}$$

Inserting the computations of A and B into (4.3.3), and then into (4.3.2), we obtain the closed form

$$\begin{aligned} D_\xi^\alpha D_x^\beta \lambda &= \sum_{k=1}^{|\beta|} (-1)^k k! i^k \sum_{\substack{\beta_1+\dots+\beta_k=\beta \\ |\beta_i| \geq 1}} c_{\beta_1, \dots, \beta_k} \sum_{\substack{\alpha' \leq \alpha \\ |\alpha'| \leq k}} \binom{\alpha}{\alpha'} \\ &\quad \times \left(\sum_{\substack{\gamma_1+\dots+\gamma_k=\alpha' \\ |\gamma_i| \leq 1}} \binom{\alpha'}{\gamma_1, \dots, \gamma_k} \prod_{i=1}^k D_\xi^{\gamma_i} ((D_x^{\beta_i} \nu) \cdot \xi) \right) \\ &\quad \times \left(\sum_{r=1}^{|\alpha-\alpha'|} \frac{(-1)^r (k+r)!/k!}{\sigma^{k+1+r}} \sum_{\substack{\mu_1+\dots+\mu_r=\alpha-\alpha' \\ |\mu_p| \geq 1}} c_{\mu_1, \dots, \mu_r} \prod_{p=1}^r (D_\xi^{\mu_p} \sigma) \right). \end{aligned} \quad (4.3.4)$$

This is the fully expanded form of $D_\xi^\alpha D_x^\beta \lambda$: a finite sum of terms, each of which is a constant times a power $\sigma^{-(k+1+r)}$ multiplied by derivatives of ν and of $\langle \xi \rangle^{2s}$.

We now bound the modulus of a single term of (4.3.4). We use:

- (i) $|\sigma| \geq \operatorname{Re} \sigma = \langle \xi \rangle^{2s}$, hence $|\sigma|^{-(k+1+r)} \leq \langle \xi \rangle^{-2s(k+1+r)}$;
- (ii) $|D_\xi^{\gamma_i} ((D_x^{\beta_i} \nu) \cdot \xi)| \lesssim \langle \xi \rangle^{1-|\gamma_i|}$, because for $\gamma_i = 0$ it is $\lesssim |\xi| \lesssim \langle \xi \rangle$ and for $|\gamma_i| = 1$ it is bounded ($\nu \in C_b^\infty$);
- (iii) $|D_\xi^{\mu_p} \sigma| \lesssim \langle \xi \rangle^{1-|\mu_p|}$, because $|D_\xi^{\mu_p} \langle \xi \rangle^{2s}| \lesssim \langle \xi \rangle^{2s-|\mu_p|} \leq \langle \xi \rangle^{1-|\mu_p|}$ (as $2s < 1$) and the linear part contributes only for $|\mu_p| \leq 1$.

Multiplying the bounds from (ii) over $i = 1, \dots, k$, and using $\gamma_1 + \dots + \gamma_k = \alpha'$,

$$\prod_{i=1}^k |D_\xi^{\gamma_i} ((D_x^{\beta_i} \nu) \cdot \xi)| \lesssim \prod_{i=1}^k \langle \xi \rangle^{1-|\gamma_i|} = \langle \xi \rangle^{k-|\alpha'|}.$$

Multiplying the bounds from (iii) over $p = 1, \dots, r$, and using $\mu_1 + \dots + \mu_r = \alpha - \alpha'$,

$$\prod_{p=1}^r |D_\xi^{\mu_p} \sigma| \lesssim \prod_{p=1}^r \langle \xi \rangle^{1-|\mu_p|} = \langle \xi \rangle^{r-|\alpha-\alpha'|} = \langle \xi \rangle^{r-|\alpha|+|\alpha'|}.$$

Therefore one term of (4.3.4) is bounded by

$$\begin{aligned} |D_\xi^\alpha D_x^\beta \lambda| &\lesssim \langle \xi \rangle^{-2s(k+1+r)} \langle \xi \rangle^{k-|\alpha'|} \langle \xi \rangle^{r-|\alpha|+|\alpha'|} \\ &= \langle \xi \rangle^{-2s(k+1+r)+k-|\alpha'|+r-|\alpha|+|\alpha'|} \\ &= \langle \xi \rangle^{-2s(k+1+r)+k+r-|\alpha|}. \end{aligned}$$

Note that the powers $|\alpha'|$ cancelled exactly. Set $\ell := k + r$. The exponent becomes

$$-2s(\ell + 1) + \ell - |\alpha| = \ell(1 - 2s) - 2s - |\alpha|.$$

The exponent $\ell(1 - 2s) - 2s - |\alpha|$ is *increasing* in ℓ , since $1 - 2s > 0$. The range of $\ell = k + r$ is bounded above: from (4.3.4), $1 \leq k \leq |\beta|$ and $1 \leq r \leq |\alpha - \alpha'| \leq |\alpha|$, hence $\ell = k + r \leq |\alpha| + |\beta|$. The maximal exponent is thus attained at $\ell = |\alpha| + |\beta|$:

$$\begin{aligned} \ell(1 - 2s) - 2s - |\alpha| &\leq (|\alpha| + |\beta|)(1 - 2s) - 2s - |\alpha| \\ &= |\alpha|(1 - 2s) + |\beta|(1 - 2s) - 2s - |\alpha| \\ &= |\alpha|[(1 - 2s) - 1] + |\beta|(1 - 2s) - 2s \\ &= -2s|\alpha| + (1 - 2s)|\beta| - 2s. \end{aligned}$$

Since (4.3.4) is a *finite* sum, summing the bounds over all terms preserves the largest exponent, and we conclude

$$|D_\xi^\alpha D_x^\beta \lambda(x, \xi)| \lesssim \langle \xi \rangle^{-2s-2s|\alpha|+(1-2s)|\beta|}$$

for all $x, \xi \in \mathbb{R}^n$ and all multi-indices α, β . This is exactly the symbol estimate of class $S_{\rho, \delta}^m$ with $m = -2s$, $\rho = 2s$, $\delta = 1 - 2s$, so $\lambda \in S_{2s, 1-2s}^{-2s}$. $\square$

4.3.2 Lemma (Adaptation of the strategy of (Epstein and Pop, 2016)). Let $s \in (\frac{1}{4}, \frac{1}{2})$ and let λ be the candidate parametrix symbol of Lemma 4.3.1. Then

$$T_\lambda \circ \mathcal{L} = I + T_v,$$

where $v \in S_{2s, 1-2s}^{1-4s}(\mathbb{R}^n \times \mathbb{R}^n)$. In particular, since $s > \frac{1}{4}$, the order $1 - 4s$ is negative, so T_v has negative order and $I + T_v$ is an elliptic pseudo-differential operator with symbol in $S_{2s, 1-2s}^0(\mathbb{R}^n \times \mathbb{R}^n)$.

Proof of Lemma 4.3.2. By Lemma 4.3.1, the candidate parametrix symbol $\lambda = 1/\sigma$ satisfies $\lambda \in S_{2s, 1-2s}^{-2s}$, while the symbol σ of $\mathcal{L}$ belongs to $S_{1,0}^1 \subset S_{2s, 1-2s}^1$ by Theorem 4.1.1.

Since $s > \frac{1}{4}$, we have

$$\delta = 1 - 2s < 2s = \rho,$$

so, by Remark 3.6.4, the composition theorem (Theorem 3.4.3) is valid in the class $S_{2s, 1-2s}$. Hence $T_\lambda \circ \mathcal{L} = T_c$ is a pseudo-differential operator whose symbol c belongs to $S_{2s, 1-2s}^{1-2s}$ and admits the asymptotic expansion

$$c(x, \xi) \sim \sum_{\alpha \geq 0} \frac{(-i)^{|\alpha|}}{\alpha!} (\partial_\xi^\alpha \lambda)(x, \xi) (\partial_x^\alpha \sigma)(x, \xi), \quad (4.3.5)$$

in the sense that, for every positive integer N ,

$$c - \sum_{|\alpha| < N} \frac{(-i)^{|\alpha|}}{\alpha!} (\partial_\xi^\alpha \lambda)(\partial_x^\alpha \sigma) \in S_{2s, 1-2s}^{1-2s-(\rho-\delta)N}.$$

We identify the terms of (4.3.5).

The leading term $\alpha = 0$. Since σ never vanishes (Lemma 4.3.1), the term $\alpha = 0$ is simply

$$c_0(x, \xi) = \lambda(x, \xi) \sigma(x, \xi) = \frac{1}{\sigma(x, \xi)} \sigma(x, \xi) = 1.$$

The terms of order $|\alpha| \geq 1$. Fix a multi-index α with $|\alpha| \geq 1$. By Lemma 4.3.1, each ξ -derivative of λ lowers the order by $\rho = 2s$, so

$$\partial_\xi^\alpha \lambda \in S_{2s, 1-2s}^{-2s-2s|\alpha|}.$$

On the other hand $\sigma \in S_{1,0}^1$, and an x -derivative does not raise the order of a symbol; viewing σ in the larger class,

$$\partial_x^\alpha \sigma \in S_{1,0}^1 \subset S_{2s, 1-2s}^1.$$

Since the product of two symbols adds their orders,

$$(\partial_\xi^\alpha \lambda)(\partial_x^\alpha \sigma) \in S_{2s, 1-2s}^{1-2s-2s|\alpha|} \subseteq S_{2s, 1-2s}^{1-4s},$$

the inclusion holding because $|\alpha| \geq 1$ gives the order bound $1 - 2s - 2s|\alpha| \leq 1 - 2s - 2s = 1 - 4s$.

Since $c_0 = 1$, the term $\alpha = 0$ contributes nothing to v , and

$$v(x, \xi) = \sum_{|\alpha| \geq 1} \frac{(-i)^{|\alpha|}}{\alpha!} (\partial_\xi^\alpha \lambda)(\partial_x^\alpha \sigma).$$

Each term of this series lies in $S_{2s, 1-2s}^{1-4s}$ by the previous step; hence, by the asymptotic summation theorem (Theorem 3.3.2, valid in $S_{2s, 1-2s}$ by Remark 3.6.4), the series sums to a symbol in $S_{2s, 1-2s}^{1-4s}$. Therefore

$$v \in S_{2s, 1-2s}^{1-4s}(\mathbb{R}^n \times \mathbb{R}^n), \quad T_\lambda \circ \mathcal{L} = T_c = T_{1+v} = I + T_v.$$

Finally, the hypothesis $s > \frac{1}{4}$ gives $1 - 4s < 0$: the remainder symbol v has *strictly negative* order. Consequently $|v(x, \xi)| \lesssim \langle \xi \rangle^{1-4s} \rightarrow 0$ as $|\xi| \rightarrow \infty$, so there is $R > 0$ with $|v(x, \xi)| \leq \frac{1}{2}$ for $|\xi| \geq R$, whence

$$|1 + v(x, \xi)| \geq 1 - |v(x, \xi)| \geq \frac{1}{2}, \quad |\xi| \geq R.$$

This is the ellipticity estimate of order 0. Thus $1 + v \in S_{2s, 1-2s}^0$ is elliptic and $I + T_v$ is an elliptic pseudo-differential operator, as claimed. $\square$

The two-stage construction below is modelled on Epstein and Pop (2016); the proofs use the calculus of Chapter 3 and Lemmas 4.3.1-4.3.2 established above.

4.3.3 Theorem. Let $s \in (\frac{1}{4}, \frac{1}{2})$ and let $\nu \in C_b^\infty(\mathbb{R}^n; \mathbb{R}^n)$. Then the pseudo-differential operator $\mathcal{L}$ admits a two-sided parametrix T_Λ , with symbol $\Lambda \in S_{2s, 1-2s}^{-2s}(\mathbb{R}^n \times \mathbb{R}^n)$, that is

$$T_\Lambda \mathcal{L} = I + T_{R_1}, \quad \mathcal{L} T_\Lambda = I + T_{R_2},$$

where $R_1, R_2 \in S^{-\infty}$.

Proof. By Lemma 4.3.2, since $s > \frac{1}{4}$,

$$T_\lambda \circ \mathcal{L} = T_c, \quad c = 1 + v, \quad v \in S_{2s, 1-2s}^{1-4s},$$

and the symbol $c = 1 + v$ is elliptic of order 0 in $S_{2s, 1-2s}^0$. By the elliptic parametrix theorem (Theorem 3.6.2), valid in the class $S_{2s, 1-2s}$ because $\delta = 1 - 2s < 2s = \rho$ (Remark 3.6.4), the elliptic operator T_c admits a two-sided parametrix $T_{c'}$ with $c' \in S_{2s, 1-2s}^0$, namely

$$T_{c'} \circ T_c = I + T_{R_1}, \quad T_c \circ T_{c'} = I + T_{R_2}, \quad R_1, R_2 \in S^{-\infty}. \quad (4.3.6)$$

Define

$$T_\Lambda := T_{c'} \circ T_\lambda.$$

Then, using associativity of composition and $T_\lambda \circ \mathcal{L} = T_c$,

$$T_\Lambda \mathcal{L} = T_{c'} \circ (T_\lambda \circ \mathcal{L}) = T_{c'} \circ T_c = I + T_{R_1},$$

with $R_1 \in S^{-\infty}$ by (4.3.6). Thus T_Λ is a left parametrix of $\mathcal{L}$.

Moreover, $\lambda \in S_{2s, 1-2s}^{-2s}$ (Lemma 4.3.1) and $c' \in S_{2s, 1-2s}^0$, so by the composition theorem (Theorem 3.4.3) the symbol Λ of $T_\Lambda = T_{c'} \circ T_\lambda$ belongs to $S_{2s, 1-2s}^{0+(-2s)} = S_{2s, 1-2s}^{-2s}$.

By Proposition 4.2.1, the symbol of $\mathcal{L}^*$

$$\sigma^*(x, \xi) = \langle \xi \rangle^{2s} - i \nu(x) \cdot \xi - \operatorname{div} \nu(x) + u(x, \xi),$$

with $u \in S_{1,0}^{-1}(\mathbb{R}^n)$. In particular, the principal symbol of $\mathcal{L}^*$ is:

$$\langle \xi \rangle^{2s} - i \nu(x) \cdot \xi,$$

which differs from the principal symbol of $\mathcal{L}$ only by the sign of the imaginary part. Consequently, the construction of $\mathcal{L}$ applies to $\mathcal{L}^*$ (is whose directional ellipticity is identical). Hence there exists $M \in S_{2s, 1-2s}^{-2s}$ such that

$$T_M \mathcal{L}^* = I + T_{S_1}, \quad S_1 \in S^{-\infty}.$$

Taking adjoints and using that adjunction preserves the Hörmander class $S_{2s, 1-2s}^{-2s}$ when $\delta < \rho$ (here $1 - 2s < 2s$), we obtain

$$\mathcal{L} T_{M^*} = I + T_{S_1^*}.$$

Since $S^{-\infty}$ is stable under adjunction, $S_1^* \in S^{-\infty}$. Setting $\Lambda' := M^*$, we arrive at

$$\mathcal{L} T_{\Lambda'} = I + T_{R_2}, \quad R_2 \in S^{-\infty}. \quad (4.3.7)$$

Since $R_1, R_2 \in S^{-\infty}$ and $S^{-\infty}$ is a two-sided ideal under composition, hence $T_\Lambda = T_{\Lambda'}$ modulo $\operatorname{OP} S^{-\infty}$. Therefore T_Λ is simultaneously a left and a right parametrix:

$$T_\Lambda \mathcal{L} = I + T_{R_1}, \quad \mathcal{L} T_\Lambda = I + T_{R_2}, \quad R_1, R_2 \in S^{-\infty},$$

with $\Lambda \in S_{2s, 1-2s}^{-2s}$. □

4.4 Discussion

The analysis carried out in this chapter establishes a precise symbolic description of the fractional Bessel operator with drift,

$$\mathcal{L} = (I - \Delta)^s + \nu(x) \cdot \nabla, \quad s \in (0, 1/2),$$

here, where the dominant first-order term destroys the global ellipticity of the symbol (the supercritical regime).

First, on the *symbolic structure*: Theorem 4.1.1 shows that $\mathcal{L}$ is a pseudo-differential operator associated to the symbol given by:

$$\sigma(x, \xi) = \langle \xi \rangle^{2s} + i \nu(x) \cdot \xi \in S^1(\mathbb{R}^n \times \mathbb{R}^n), \quad n \geq 2,$$

of order exactly 1 as soon as the drift does not vanish identically.

Second, on the *ellipticity properties*: Proposition 4.1.2 shows that, for $n \geq 2$, the symbol σ is not elliptic in the classical sense: along the characteristic hyperplane $\Sigma_x = \{\xi : \nu(x) \cdot \xi = 0\}$ the modulus $|\sigma|$ drops to order $2s < 1$. Lemma 4.1.3 refines this picture by showing that, away from Σ_x , the symbol remains elliptic of order 1 on every closed cone $\Gamma_{x,\theta}$.

Third, on the *construction of a parametrix and the formal adjoint*: Theorem 4.3.3 provides a two-sided parametrix of $\mathcal{L}$ in the Hörmander class $S_{2s, 1-2s}^{-2s}$ for $s \in (1/4, 1/2)$, with regularising remainder. Proposition 4.2.1 computes the formal L^2 -adjoint $\mathcal{L}^*$ explicitly, with principal symbol

$$\sigma^*(x, \xi) = \langle \xi \rangle^{2s} - i \nu(x) \cdot \xi,$$

The construction of the parametrix is sharp within the symbolic framework adopted here. The condition $s > 1/4$ is dictated by the structural constraint of the Hörmander calculus in $S_{\rho,\delta}^m$: the composition theorem produces a remainder of strictly decreasing order only when $\delta < \rho$, which in our case reads $1 - 2s < 2s$. A second limitation concerns the regularity of the drift. The hypothesis $\nu \in C_b^\infty(\mathbb{R}^n; \mathbb{R}^n)$ is essential to the symbol estimates throughout the chapter: the bounds on the derivatives of ν control directly the constants in the class S^1 .

It is worth emphasising that the supercritical regime $s \in (0, 1/2)$ is genuinely different from the subcritical one $s \in (1/2, 1)$, where the diffusion $(I - \Delta)^s$ dominates and the operator inherits the ellipticity of the Bessel potential. The threshold $s = 1/2$ is the value at which the diffusion and the drift contribute to the same order in the symbol; below this value, the drift dominates at high frequencies and the loss of ellipticity is unavoidable.

The strategy of the parametrix construction in Section 4.3 is inspired by Epstein and Pop (2016), who treat the supercritical fractional Laplacian $(-\Delta)^s + \nu(x) \cdot \nabla$. The present work departs from theirs in several ways. First, the operator $(I - \Delta)^s$ replaces $(-\Delta)^s$, removing the low-frequency singularity of the pure fractional Laplacian and making the symbol globally well-defined on $\mathbb{R}^n \times \mathbb{R}^n$. The directional ellipticity of Lemma 4.1.3 and the ellipticity estimate on the cone $\Gamma_{x,\theta}$ is absent from Epstein and Pop (2016). These results, together with the explicit computation of the formal adjoint in Proposition 4.2.1, constitute the original contributions of this work; the parametrix construction of Lemmas 4.3.1-4.3.2 and Theorem 4.3.3 uses the strategy of Epstein and Pop (2016) but is carried out entirely within the calculus developed in Chapter 3.

5. Conclusion

This dissertation has given a self-contained account of the theory of pseudo-differential operators on $\mathbb{R}^n$. It has moved, step by step, from the analytical groundwork to a non-trivial application.

Chapter 2 gathered the analytical tools we would need. These were multi-index calculus, differentiation under the integral sign, convolution, the Fourier transform on the Schwartz space $\mathcal{S}(\mathbb{R}^n)$, tempered distributions, and the dyadic partition of unity together with Taylor's formula with integral remainder. All of this material is classical. We still proved it in full, because it is used again and again in the symbolic calculus that follows.

Chapter 3 built the core calculus of pseudo-differential operators in the standard Kohn–Nirenberg class $S^m = S_{1,0}^m$. We established four basic results, each with a complete proof. First, the asymptotic summation theorem, which gives a meaning to formal symbol expansions. Second, the composition formula, which shows that the product of two pseudo-differential operators is again such an operator, and provides an explicit asymptotic symbol for it. Third, the construction of the formal adjoint. Fourth, the existence of a two-sided parametrix for elliptic operators.

Chapter 4 put this machinery to work on a problem that lies outside the classical elliptic theory: the fractional Bessel potential with drift,

$$\mathcal{L} = (I - \Delta)^s + \nu(x) \cdot \nabla, \quad s \in \left(0, \frac{1}{2}\right),$$

in the supercritical regime. Here the first-order drift dominates the diffusion at high frequencies. We showed that $\mathcal{L}$ is a pseudo-differential operator of order one whose symbol is not elliptic once $n \geq 2$: ellipticity breaks down along the characteristic hyperplane $\Sigma_x = \{\xi : \nu(x) \cdot \xi = 0\}$. The key observation was that ellipticity nonetheless survives *microlocally*. On every conic region $\Gamma_{x,\theta}$ kept away from Σ_x a directional form of ellipticity holds, and this reflects the geometry of the drift. Working from this estimate, and passing to the Hörmander class $S_{2s,1-2s}^{-2s}$, we constructed an explicit two-sided parametrix for $\tilde{T}$ in the sub-regime $s \in (1/4, 1/2)$, and we computed its formal adjoint.

Several questions remain open, and they arise naturally from the present work. The complementary range $s \in (0, 1/4]$ is the most immediate. To reach it, one would need either a microlocal decomposition fitted to the cone $\Gamma_{x,\theta}$ and the hyperplane Σ_x , or a local straightening of the drift in the spirit of [Epstein and Pop \(2016\)](#). Either route would carry the results of this dissertation across to the full supercritical regime.

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Appendix A. Proofs of technical result

A.1 The measurable functional calculus

In the proof of Theorem 4.1.1 we defined the fractional Bessel potential $(I - \Delta)^s$ through the spectral (measurable) functional calculus, writing

$$((I - \Delta)^s u)^\wedge(\xi) = (1 + |\xi|^2)^s \hat{u}(\xi).$$

We collect here the theorem that justifies this construction. It states that any measurable functional calculus is, up to a unitary equivalence, a multiplication operator. We follow Haase (2006); the statement is reproduced with the notation adapted to the present work.

A.1.1 Theorem (Multiplication representation of a measurable calculus). *Let (Φ, H) be a measurable calculus on the measurable space (X, Σ) . Then there exist a semi-finite measure space $(\Omega, \mathcal{F}, \mu)$, a unitary operator $U : H \rightarrow L^2(\Omega, \mathcal{F}, \mu)$, and an injective $*$ -homomorphism $T : \mathcal{M}(X, \Sigma) \rightarrow \mathcal{M}(\Omega, \mathcal{F})$ such that*

$$\Phi(f) = U M_{Tf} U^{-1} \quad \text{for all } f \in \mathcal{M}(X, \Sigma), \quad (\text{A.1.1})$$

where M_{Tf} denotes multiplication by Tf . Moreover, Φ is continuous with respect to pointwise convergence of sequences.

Proof. We only sketch it, full details being available in standard references, e.g. (Haase, 2006, Ch. VII) and the appendices therein.

For each $x \in H$, let μ_x be the measure on (X, Σ) defined by

$$\mu_x(A) = \langle \Phi(\mathbf{1}_A)x, x \rangle, \quad A \in \Sigma.$$

Then

$$\langle \Phi(f)x, x \rangle = \int_X f d\mu_x, \quad f \in \mathcal{M}_b(X, \Sigma).$$

Define

$$Z_x := \overline{\{\Phi(f)x : f \in \mathcal{M}_b(X, \Sigma)\}}.$$

Then the map $\mathcal{M}_b(X, \Sigma) \rightarrow Z_x$, $f \mapsto \Phi(f)x$, is isometric with respect to $\|\cdot\|_{L^2(\mu_x)}$ and hence extends to a unitary operator

$$V_x : L^2(X, \Sigma, \mu_x) \longrightarrow Z_x.$$

Z_x and $Z_x^\perp$ are $\Phi(\mathcal{M}_b(X, \Sigma))$ -invariant. Employing Zorn's lemma, one finds a maximal set $(x_\alpha)_\alpha$ of unit vectors $x_\alpha \in H$ such that the spaces Z_{x_α} are pairwise orthogonal and satisfy

$$H = \ell^2\text{-}\bigoplus_{\alpha} Z_{x_\alpha}.$$

For each α let $X_\alpha := X \times \{\alpha\}$ be a copy of X , so that the X_α are pairwise disjoint. Let $\Omega := \bigsqcup_{\alpha} X_\alpha$ be their disjoint union, let $\mathcal{F}$ be the largest σ -algebra on Ω such that all inclusion maps $X_\alpha \rightarrow \Omega$ are measurable, and let μ be the measure on $\mathcal{F}$ defined by

$$\mu(B) := \sum_{\alpha} \mu_{x_\alpha}(B \cap X_\alpha), \quad B \in \mathcal{F},$$

and let $K := L^2(\Omega, \mathcal{F}, \mu)$. Define the unitary operator

$$U : H \rightarrow K \quad \text{such that} \quad U^{-1} = V_{x_\alpha} \text{ on } L^2(X_\alpha, \Sigma, \mu_{x_\alpha}) \subseteq K.$$

Then define the mapping $T : \mathcal{M}(X, \Sigma) \rightarrow \mathcal{M}(\Omega, \mathcal{F})$ by

$$(Tf)(x, \alpha) := f(x), \quad x \in X.$$

Then T has the desired properties. Define the measurable calculus (Ψ, K) by $\Psi(f) := M_{Tf}$ (that is, Ψ is the pull-back of the multiplication operator calculus by T). Then, by construction, the identity

$$M_{Tf} = U \Phi(f) U^{-1}$$

holds for all $f \in \mathcal{M}_b(X, \Sigma)$. Consequently, by the continuity remarks above, it must hold for all $f \in \mathcal{M}(X, \Sigma)$, which is (A.1.1). $\square$

A.1.2 Remark (Application to $(I - \Delta)^s$). The Bessel potential used in Theorem 4.1.1 is the special case of Theorem A.1.1 in which $H = L^2(\mathbb{R}^n)$, the unitary U is the Fourier transform $\mathcal{F}$ (Theorem 2.5.6), and Φ is the functional calculus of the self-adjoint operator $I - \Delta$. Under $\mathcal{F}$, the operator $I - \Delta$ becomes multiplication by the symbol $1 + |\xi|^2$, so for any measurable $f : [1, \infty) \rightarrow \mathbb{C}$,

$$f(I - \Delta) = \mathcal{F}^{-1} M_{f(1+|\xi|^2)} \mathcal{F}.$$

Taking $f(t) = t^s$ gives precisely

$$(I - \Delta)^s u = \mathcal{F}^{-1} [(1 + |\xi|^2)^s \widehat{u}],$$

which is the identity (4.1.1) on which the symbolic analysis of Chapter 4 rests. In particular, since $1 + |\xi|^2 \geq 1$, the function $t \mapsto t^s$ is bounded on the spectrum of $I - \Delta$ for $s \leq 0$ and of polynomial growth for $s > 0$, so the calculus is well defined on $\mathcal{S}(\mathbb{R}^n)$.